



Master Thesis

MSc Finance and Investments

**How do country-level regulatory and political factors
influence M&A performance?
Evidence from the infrastructure sector**

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Preface

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Abstract

In this paper, we analyzed the effects of regulatory and political factor levels on the M&A performance in the infrastructure sector. The analysis of 1,106 worldwide (out of which 310 cross-border) deals builds upon the previous research on investors' protection effects (La Porta et al., 1997) and spillover by law effect (Martynova & Renneboog, 2008) and aims at filling in the gap in research focused on this industry. Furthermore, previously unused or rarely used variables are employed in the regressions. Firstly, we analyze the influence of World Bank's regulatory and political index levels on the acquirer's CARs upon the transaction announcement. Secondly, we investigate deeper the influence of acquirer and target country differences in these factors of CARs. Thirdly, we examine CARs with respect to the investors' protection level in the target country and target company public status. Our results support the positive impact of the superior regulatory environment (apart from the investors' protection) on acquirer's CARs, especially in diversifying acquisitions, and moderately positive impact of such a political environment. Yet, the analysis of differences in the factor levels supports the positive spillover by law hypothesis implying value-creating effect upon the announcement coming from the possibility of a lower-ranked target adopting better corporate governance standards post-acquisition. Furthermore, our results suggest that acquirers of private companies in countries with lower investors' protection levels benefit from significantly higher positive CARs, potentially thanks to an additional discount apart from the liquidity one.

Keywords: M&A, M&A performance, infrastructure, political factors, regulatory factors

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1. Introduction

Mergers & Acquisition (M&A) activities have exhibited a record-breaking volume in recent years, with the overall volume in the year 2018 of \$3.35tn according to the Dealogic financial data platform. M&A intensively reshapes the business landscape as it became a regular business activity to support quickly geographical growth (Gaughan, 2007) or exploit the possibility of economies of scale (Gaughan, 2002). The infrastructure sector has been following the intensity of the M&A activity of other industries with the aggregate deals' value of \$916bn completed globally in 2017 based on the 2018 Prequin Global Infrastructure Report. This has been triggered by market openness to privatization and recent liberalization (Monastyrenko, 2017). The increased interest by investors due to the specific asset class characteristics, however, does not come without a downside. Management teams and investors often present the political and regulatory environment as the major potential stream of risk for their returns, operations or M&A success (Altman, 2005).

Research investigating the M&A performance and M&A determinants has been extensive, however, regulatory, legal or political aspects were mostly research in relation to the corporate governance standards, the results are inconclusive or differ to a large extent due to specific deal characteristics (see, e.g. Stark & Wei, 2004; John et al., 2010) and methodology to proxy for such effects evolves. Legal and regulatory factors, such as investor protection, are mentioned often and linked to stock markets development (La Porta et al., 1997), firm valuation (La Porta et al., 2002), as well as spillover by law effects (Martynova & Renneboog, 2008). Scholars research both shareholder right basis (common vs. civil law type, La Porta et al., 1998), as well as creditor rights (Djankov et al., 2007), showing higher cross-country M&A activity in case of better acquirer shareholder protection (Rossi & Volpin, 2004) and decrease in returns when creditors' protection is lower (John et al., 2010). Subsequently, researchers studied market attractiveness depending on other related factors such as regulatory, economic or political environment maturity (Moeller et al., 2016). Factors such as control of corruption or protection of investors proved to have an influence on the M&A performance (see, e.g. Ellis et al. 2011; Martynova & Renneboog, 2008) and this topic still remains relevant.

The infrastructure sector (comprising utilities, energy and transport, and telecommunications infrastructure) has become of concern to researchers, as well as non-academics especially following the larger wave of privatization and deregulation in Europe in the 1990s and beginning of 21st century. The industry can be characterized by its economic importance, as it

contributes to long-term growth boost and thanks to the economies of scale, the growth exceeds the capital stock (Balász et al., 2009). This aspect, as well as the cyclical nature of the industry causes its close observation by regulators specifying e.g. maximum rate of return. Other specific characteristics of this asset class such as economies of scale and scope, natural monopoly, stable periodic cash flows or low human capital risk (Bartunek et al., 1993) also drew the interest of investors, encouraging the surge in M&A deals as well as their value on a significant scale (Datta et al., 2013). Furthermore, specifically \$3.7 trillion a year by 2035 should be deployed in the infrastructure sector worldwide to maintain economic growth according to McKinsey Global Institute (2017). This need for bridging the infrastructure funding gap, ongoing sector privatization, as well as the increasing popularity by investors to include this asset class in their portfolio (e.g. to limit risk), motivates us to investigate the drivers of M&A performance in more detail. Due to the regulated nature of the industry and types of risks related to acquisitions of companies in this sector, regulatory and political factors or maturity require naturally special awareness in order to understand the magnitude of their impact. Infrastructure is typical by the rate case regulation, where the total revenue requirement is based on the rate base (value of the company's assets), allowed rate of return on them and the expenses. The trade-off arises due to the exclusive right to distribute electricity. Therefore, the amount that can be charged and the profit margin is regulated. Experts in the field emphasize e.g. in World Bank's Reforming Infrastructure policy research report (2004) that the efficiency of regulation in terms of the predictability, accountability and transparency is necessary to sustain the investors' interest and financial deployment in the sector which is ultimately leading to performance, productivity, coverage improvement, as well as provision of further shared social benefits (Kessides, 2004).

Comparing to other sectors, regulated ones exhibit also both negative returns for acquirer and positive ones for the target in the short run, however on the lower level (Datta et al., 2013). Possible explanations comprise lack of experience, concerns of the antitrust process (Ray & Thompson, 1990; Bartunek et al., 1993), as well as economic regulation (Leggio & Lie, 2000). As factors influencing the shareholder wealth creation in infrastructure M&A remain to be studied more deeply, the main research question of my thesis the following:

To what extent does the regulatory and political environment factors in the target country influence the acquirer's M&A performance in the infrastructure sector and what is the impact of differences in the target and acquirer's regulatory and political environment factors on the acquirer's M&A performance?

Our methodology follows previous research of the close sub-area of corporate governance and investors' protection by Martynova & Renneboog (2008) and John et al. (2010). A standard event study evaluating short-term M&A performance in several event-length windows to ensure the robustness of the results and assessment of possible information leakage effects provides us with acquirer's cumulative abnormal returns (CARs). These later serve as our dependent variable in the multivariate regressions diving deeper into the effect of political and regulatory environment. Our independent variables are based on carefully related indices describing the legal, corporate governance and political environment on the country level. Namely, we conduct our analysis using indices for control of corruption, political stability and regulatory quality; the two latter of those have been rarely employed. In addition, we newly introduce also the index of legal contract enforcement and a different index describing the investor's protection comparing to the standardly used La Porta et al.'s (1998). Furthermore, several control variables are employed.

The research is conducted on previously not investigated data of M&A utility, energy, telecommunications and transportation infrastructure deals (classified according to SIC industry codes) announced in period 01/01/2006 to 31/12/2017 from Thompson One database with the current status of a completed deal. Datastream database provides stock and index performance data. Regulatory and political factors data come from indices published by World Bank.

This thesis aims at filling in the gap in the narrow branch of infrastructure M&A studies and builds upon the existing research concerning finance and law launched by La Porta et al. (1998). I find that the highly ranked of regulatory and political environment indeed contributes positively to acquirer's CARs apart from the investors' protection level. The analysis of also supports the hypothesis of positive spillover by law stemming from the opportunity of lower-ranked target's improvement of corporate governance standards as a result of acquisition by a company situated in a higher-ranked country. Moreover, our results suggest that acquirers of private companies in countries with lower investors' protection levels benefit from significantly higher positive CARs, possibly thanks to an additional discount. To the best of my knowledge, although a few previously cited studies analyzed the M&A announcement returns performance, specific variables influencing their magnitude remain under-investigated, to say the least. The study also contributes to the field by examining previously non-investigated data between years 2006 and 2017 and assessing the performance of acquirers through previously unused, rarely used or differently defined variables or proxies. Better

insight into the effects of the specific regulatory and political factors contributes to the understanding of the drivers of M&A successful or failing performance which can be valuable to the academic field, investors, managers or other parties involved in the matter.

The thesis is structured in the following way: after the overview of the infrastructure industry, the M&A activities and their relations with regulatory and political factors, I formulate my hypotheses. This is followed by the methodology, models and description of the dataset used. Afterwards, we present the empirical findings, followed by their discussion and a robustness check. The paper is closed by a conclusion and a brief discussion of the study limitations, as well as possible extensions.

2. Literature review

2.1 Infrastructure Industry and M&A activity

2.1.1 *Industry overview*

The infrastructure industry managed to set itself as a separate asset class within alternative investments that continuously attracts hefty financial allocations mainly due to its characteristic traits. In the case of this particular sector, however, the investment is a major two-way street, as it also ensures primary services and functions of the economy. Thus, it provides crucial support to potential productivity growth and thriving economic development in the country.

The industry can be divided further into several specific sub-sectors, each of which also exhibits mildly different behavior with respect to the macroeconomic factors. The sub-sections that we generally understand under the infrastructure industry term comprise: a) regulated assets consisting of electric power industry, water distribution, processing, and waste systems and oil and gas midstream; b) transportation assets such as toll roads, airports and ports; c) telecommunications infrastructure including wireless and broadcast towers or satellite networks; and in some cases mentioned d) social infrastructure assets comprising school, hospitals, and other public service houses. This has not been, however, included in the previous research (e.g. Bartunek et al., 1993). Given the level of underlying risk, infrastructure real assets are (similarly to the real estate assets) classified into the categories of core, core plus, value-added and opportunistic categories (described in the order from the lowest to the highest risk exhibited). (J. P. Morgan Asset Management, 2009; Brookfield, 2016).

Despite the potentially high initial deployment of financing going hand in hand with the capital intensity of this industry, it possesses several unique traits perceived as benefits for the investors (Ammar & Eling, 2015). Firstly, the stable demand considerably insensitive or resistant to the economic cycles, as the assets provide the backbone and essential services of the well-functioning economy. This ensures the investors with stable cash flows (Bird et al., 2011). Secondly, the industry is monopolistic in nature (Finkenzeller et al., 2010) with high barriers to entry. Thirdly, the inclusion of the asset class in the portfolio provides investors with substantial benefits from diversification. Not only is the infrastructure correlated with other asset classes to a very limited level (Finkenzeller et al. 2010), diversification is also possible within the asset class by investments in the above-mentioned sub-sectors of the industry or in terms of geography. Fourthly, investors are incentivized to engage into this capital-intensive

industry by regulations employed by governments, which ensure appropriate compensation for risks and return on investment through long-term contracts (often with a length of 10 to 15 years). Finally, private-owners can also benefit from the indexation of the contracts in question with respect to the inflation or potentially other economic indicators (J. P. Morgan Asset Management), which is, however, disputed by several papers such as Rödel & Rothballer (2012) or Bird et al. (2011).

A reward does not come without risks and infrastructure sector is subject to several specific ones. Scholars also suggest that in terms of investments in infrastructure stocks, traditional asset pricing models are not applicable or capturing the risk exposure sufficiently and infrastructure-specific factors should be added to the traditional Fama-French three-factor model or others (Ammar & Eling, 2015; Wurstbauer et al., 2016). Private-owners might suffer from the unavailability of a substantial amount of return data in comparison to other asset classes due to the novelty of the asset class. In case of opportunistic assets, the construction and demand uncertainty risks appear and the capital-intensity of the asset along with the regulatory processes upon acquisitions, the liquidity of such investment is also substantially reduced unless the investor participates in an open-ended fund (J. P. Morgan Asset Management, 2009). Altman (2015) divides the risks in the infrastructure sector into endogenous and exogenous ones. Endogenous risks comprise the generally manageable risks such as performance or operating risk, whereas the exogenous risks are not inherent and comprise economic and social aspects, technological advancement risks and political and regulatory factors, which are of particular interest of ours.

Regulatory factors represent the double-edged sword in the infrastructure sector: On one hand, the close monitoring and regulations due to the monopolistic nature of the industry generally provide substantial certainty of future cash flows from the company through set rates that can be charged (e.g. given the regulatory asset base) and indexation setup in the contracts. Yet, on the other hand, the instability of the regulatory environment disincentivizes potential private owners or can have a substantial unfavorable impact both on the returns, as well as the economic development in the country that would occur, was the financial resources deployed. (Blackrock, 2015).

Political factors are closely related to the regulatory ones and there are two major types of those: geopolitical and national (Altman, 2015). On the geopolitical level, political risk comprises sanctions or anti-trust proceedings, whereas on the national one, we perceive the political risk mainly as unfavorable enactments implementations. These can be concerning

taxes, unrestored concession or renationalization in case profits are perceived as higher than adequate (Altman, 2015). Infrastructure projects are often subject to the financing of up to 30 years which poses a severe uncertainty given the conventional election period of 4 or 5 years (Blackrock, 2015). The political and regulatory risk might be, however, mitigated to a certain level by an allocation of the financial resources to specific, more favorable political, legal or regulatory regions (J. P. Morgan Asset Management, 2009). However, several cases of significant impacts due to the regulation changes could be witnessed in recent years including the Spanish retroactive tax implementation leading to the bankruptcy of over 50% of solar power companies since 2008 or disincentivization of the investors by unprecedented elimination of Levy Exemption Certificates for renewable energy in the UK in 2015. (Blackrock, 2015). On the international level, the ratification of the Kyoto Protocol obliging the countries in question with CO₂ emissions cuts (Capgemini, 2008), thus inducing shifts in the energy industry consumption, is worth mentioning. From the regulator's point of view, long-term investors may be perceived as more desired, given that short-term ones might be seeking a short-sighted profit maximization at the expense of the users of the service itself.

Recent development in the industry has been vastly determined by opening to privatization and deregulation inducing companies' attempts to restructure themselves in order to face the changing competitive landscape. M&A activity thus becomes more essential and the capacity of capital to be deployed increases. Starting in the late 20th century in the U.S. and Western European Countries, it has been spreading to Central and Eastern European region. The World Bank's Reforming Infrastructure policy research report (2004) identifies significant improvement in the performance of the sector in terms of quality, productivity and coverage thanks to the reforms introduced. Therefore, although not without its critics, private capital deployment into infrastructure can well serve as a tool to generate shared social benefits in the economy. Yet, reasonable and efficient regulation is crucial to achieving such a goal.

2.1.2 Infrastructure Industry M&A drivers

The infrastructure industry possesses several specific characteristics, which can have an impact on the M&A behavior. As such, we will address the most important ones: capital intensity, leverage, low margin and economic cycle sensitivity.

Firstly, the capital intensity of the infrastructure businesses implies that the major part of costs are fixed and variable costs account for rather minority. Such companies, therefore, can benefit more from the economies of scale arising from combination of two such assets or companies. According to Gaughan (2002), this type of synergies represents a major part of the operating

synergies. Opportunity to realize such synergies as a larger firm thus results in smaller companies to be more likely to engage in transactions.

Secondly, leverage represents an important aspect of the infrastructure M&A. In 2013, the average leverage ratio in deals was 77% according to the Prequin research. Thanks to the stability of cash flows, the companies can afford to take upon additional debt and are therefore ideal targets for leveraged buy-outs. Furthermore, leverage also generally represents a factor determining M&A behavior as according to the Jensen (1986), the unused debt capacity induces companies to engage in M&A, as debt functions as a disciplining factor on managers. The obligation to pay interest payments forces them further to accept projects with higher net present value.

Thirdly, due to the capital intensity of the companies in this sector, lower profit margin is expected. Since the utility sector is a widely needed service, the regulation imposed on it reduces the profitability that can be achieved. Yet, for example with post-acquisition rearrangement of the capital structure which is one of the determinants of the allowed rate of return, profitability may be increased. This would be in line with the proposition that targets with lower profitability are more likely to be attractive for acquirers (Sorensen, 2000).

Fourthly, due to the provision of the economy backbone, the infrastructure sector is also sensitive to the economic cycle. Such quality comes with the fact that increasing usage of e.g. electricity during the economic boom periods increases the attractiveness of the sector, yet also represents a risk exposure when going into crisis. As companies need to face such changes, M&A might be induced as suggested by Mitchell & Mulherin (1996). The thriving periods themselves also offer stronger capital markets and optimistic attitude with regards to the future prospects (Gregoriou, 2007), leading to increasing transaction activity. Facing the risk of counter-cyclical deadweight losses through diversifying mergers has been also discussed in the related literature (Berger & Ofek, 1995).

2.1.3 Infrastructure Industry Mergers & Acquisitions

Deregulation as a structural industry shock has been found to have a significant effect on the increased M&A activity level by Mitchell & Mulherin (1996) which lead academic researchers to assess and understand the outcome in more detail. Apart from wealth and value creation for the shareholders of the companies involved, subsequent technology level investment (found to be increased by Majumdar et al., 2014), eco-efficiency level impact (decreasing in case of

cross-border M&A by Monastyrenko, 2017) or productive efficiency (Saastamoinen et al., 2017) has been of interest. Our focus will be on the first aspect mentioned.

As the infrastructure sector itself, even the M&A activities within the sector appear to demonstrate specific traits (Berry, 2000). The risk of human capital diversification through mergers (Amihud & Lev, 1981; Shleifer & Vishny, 1989) can be reduced thanks to the stability of the companies' performance in this sector (Bartunek et al., 1993). Other specific factor for the infrastructure M&A presented by Bartunek et al. (1993) is the idea that due to the high cash obligations in the form of regular dividends, that manager engages in value-destroying mergers that are not in favor of the shareholders (Bradley et al., 1988) and exhibit inefficient use of resources suggested by Jensen (1986). However, in light to the recent development in the reduction of dividend payouts, this idea might not hold anymore. Additionally, severe regulatory scrutiny lead to a more complicated acquisition process (Studness, 1989). Potentially of the most importance, due to the conventional regulation of the rate of return for the companies, the shareholder value creation might be limited, as the merger-related value enhancement is passed on to the end-users through lower rates (Norris, 1990). As such, Bartunek et al. (1993) suggested the expected reaction upon acquisition to be positive but lower in comparison to the non-regulated industries, which was found to be consistent with the results of their study. They further find a negative effect on the acquirer shareholder's wealth, however, the combined effect of the acquisition was positive. A contrary result with respect to the combined effect has been found by Berry (2000), despite the standalone effects having the same sign and comparable to the previous study. The recent substantial surge in the European utility M&As following the deregulation was also researched by Datta et al. (2013) who focused also on the long-term performance of the M&As undertaken between the years 1990 and 2006. Although the results of lower acquirer short-term losses corroborated some of the previous research, they also found combined losses in the long-term which might pose a negative incentive towards the investors in this sector.

In terms of specific explanatory variables affecting the M&A value creation upon announcement, Bartunek et al. (1993) find the diversifying mergers as yielding less negative results for the bidder and significantly more positive results in terms of CARs for the target and the business combined. However, it should be noted that the statistical inference might suffer from a substantially small dataset employed in the analysis. Their analysis suggests that bidder suffers more in the full acquisition (-1.7%), target enjoys significantly higher CARs (10-18%) in this setting and the acquisition is value creating overall, which is, however, disputed

by the opposite later results of Leggio and Lien (2000). These also provide evidence of regulation affecting the M&A performance with significantly negative abnormal returns of the acquiring firms in the period post-1992 when the Energy Policy Act was implemented. Uncertainty about the regulatory environment is further suggested by Berry (2000) as the reason for a reluctance to pay a high premium in the cross-border utility M&As. They are found as having no significant negative wealth effects for the shareholders of the acquirers in comparison with the M&As undertaken domestically. The above-summarized analyzes suggest that the obtained results are not unanimous and further analysis of other factors influencing the value creation, as well as their preservation over time is to be carried out.

2.2 M&A Activity and its relation to regulatory and political aspects

M&A has been a subject of extensive academic research. This chapter aims at briefly summarize the most important topics of this area in relation to the focus of the thesis: (1) Motives for the M&A activity; (2) value creation at M&A and its determinants; and (3) relation of the regulatory and political factors to the M&A area.

2.2.1 M&A Activity Motives

Determinants for undertaking the M&A transaction have been largely analyzed in order to understand better the underlying motives, a particular attractiveness of a specific market (Moeller et al., 2016) or potential influence on the success of the deal itself (Calipha et al., 2010). Within the broad landscape of the motives suggested by the scholars, we can primarily distinguish between rational, in other words, economic reasons and the irrational ones, driven for instance by behavioral biases and market imperfections.

In terms of rational motives for M&A activities, possible synergies belong to the most mentioned ones across the industries. Exploitation of economies of scope and scale may lead to financial, managerial and primarily operating synergies through cost reduction (Gaughan, 2002) throughout the whole company's supply chain and operations. Closely related to the synergy motive, we observe resource-driven acquisition be it in form of financial resources due to the acquirer's lower cost of capital following the merger (Trautwein, 1990), or in the form of intellectual property and strategic acquisition of specific asset which might lead to the increased competitive advantage (Wang & Boateng, 2007). Facing the competitive landscape successfully due to the post-M&A organizational learning (Vermeulen & Barkema, 2001) may be an important merger determinant especially for developing economies. With these being main targets, M&A serve also as a tool for market share or foreign market acquisition due to

the relative ease, high success rate and speed of the process (Gaughan, 2007) comparing to, for instance a brand-new company establishment or a construction with a necessity of a regulatory approval (McLaughlin & Mehran, 1995). Thanks to such transaction, the risk of potentially costly inconveniences stemming from the cultural or institutional differences might be reduced (Barkema et al., 1996). On the other hand, firms might intentionally search for differences, e.g. in terms of industry and geography through diversifying mergers. Such deals enable companies reduction of e.g. bankruptcy risk (Mukhejee et al., 2003), as well as the exploitation of e.g. tax optimization through international structures (Morck & Yeung, 1992). Furthermore, companies undertake M&As on an increased level due to the industry shocks in the form of deregulation as described by Mitchell and Mulherin (1996). Additionally, tax benefits such as tax loss transfers or tax expense reductions through increased depreciation following the transaction are mentioned as a low-level, rather supportive motivational factor by Auerbach and Reihus (1987).

Within the area of irrational motives for undertaking M&A transactions, one of the central problems of corporate finance in general, the agency problem can be found. Given that the interests of shareholders and managers often do not coincide, M&A activity may occur as a result of the manager's own will using the decision-making power he possesses (Shleifer & Vishny, 1988). The performance incentives in terms of compensation structures related to the company size thus also motivate the manager into empire-building behavior. This might come at a costly price for the shareholders given the possibility of engagement into mergers (projects) which are not value-maximizing as presented in the cash-flow theory by Jensen (1986). Furthermore, the hubris theory by Roll (1986) suggests that management with outstanding performance in the years prior to the M&A is inclined to overpay for the target company. The CEO's initially perceived the target as undervalued given his assumption of a possible increase in the target's value under his management. Increased appetite for M&A activity as a result of management overconfidence was suggested by the findings of Malmendier & Tate (2004). Their results also provide evidence that overconfidence induces M&A transactions of lower quality when significant financial resources are available in the company.

2.2.2 M&A performance

Scholars have extensively analyzed the M&A performance and value creation with inconclusive results and a plethora of possible explanatory variables upon which the success or failure of the deal is often dependent. Assessment of such an event faces inherently the problem of possible uniqueness of each deal as suggested by Lubatkin (1987) or its endogeneity

accounted only rarely in academic studies (see, e.g. Savor & Lu, 2009). Academic research generally considers M&A to be value-creating, as it is the target who significantly gains from the acquisition and the acquirer's returns incurs no or unsubstantial gains or losses (see, e.g. Jensen & Ruback, 1983; Moeller & Schlingemann, 2005; Andrade et al., 2001) in terms of the short-term. This is also the horizon on which we are going to focus in the following section for the sake of brevity. Following the intuition of the framework suggested by Halebian et al. (2009) classifying the variables influencing the performance into antecedents, in other words, motivational factors for the transaction itself, and the internal and external moderators of the deal, an overview (by far non-exhaustive) is presented.

Rational motives for acquisitions yield varying results for the acquiring company. Foreign market acquisitions through cross-border mergers result in lower performance comparing to domestic M&A in most studies regardless of the measures (Moeller et al., 2005). A comprehensive study by Moeller & Schlingemann (2005) concludes that the acquirer benefits from lower returns in case of acquisitions of foreign market companies comparing to those realized in M&As undertaken domestically, although the findings, for instance Doukas & Travlos (1988) suggest otherwise. Also the ambition of diversity through mergers is generally confronted by negative returns realized by the acquirer (Berger and Ofek, 1995). Morck et al. (1990) suggest that a diversifying deal is value-destroying as a result of the manager's pursuit of empire building and personal interests. Thus, the researchers connect the rational and irrational M&A motives.

Irrational motives also appear to be value destroying namely for the acquirer's shareholders. Managerial empire building associated with the retention of excessive cash amount in the company (Jensen, 1986) chastises the acquiring company later upon the acquisition announcement as the market prices in the inclination of the manager towards projects with a negative present value. Jensen (1986) provides evidence of a 2% acquirer's share price decrease upon the announcements. Harford (1999) further presents findings of negative cumulative abnormal stock returns, as well as decreasing operating performance in relation to the free cash flow reduction due to the merger. Additionally, negative acquirer's returns upon announcement are also explained by Roll's (1986) hubris theory and overconfidence motives supported by the findings of e.g. Malmendier & Tate (2004). Using option exercise as a proxy for overconfidence measure, the results suggest a higher tendency for undertaking (even sub-par) mergers which are then reflected by 90 basis point decreasing market reaction.

Specific deal characteristics have proven to influence on the M&A outcome. Firstly, the method of payment determined the returns upon announcement. Cash deals yield positive returns comparing to negative ones in case of stock deals (Loughran & Vijh, 1997) as a result of the signaling and acquirer's overvaluation in the latter case (Myers & Majluf, 1984). Secondly, acquisitions of private companies were empirically found to yield better results for the bidder, primarily thanks to the liquidity premium, whereas public firms bring significant losses (Moeller et al., 2004). Finally, competition level in the process (Bradley et al., 1988), deal attitude (Lang et al., 1989) or size (Alexandridis et al., 2013) also influence the deal outcome among others.

In terms of cross-border M&As, the country-level factors may also explain differing performance results. GDP growth in the host (target) country can act as a catalysator for the success of the deal (Evenet, 2002). Geographical proximity as suggested by e.g. Martynova and Renneboog (2008), historical ties (Chowdhury & Maung, 2018) or cultural relatedness can be a determinant of undertaking the deal, as well as its performance. For instance, a prominent theory was suggested by Hofstede and his cultural dimension indices (Hofstede, 1983). Aspects included in the index, such as individualism or long-term orientation may affect the M&A process decisions and diligence put into such endeavor.

Similarly to Hofstede (1983), scholars (see e.g. Moeller et al., 2016) also analyzed variables or constructed indices proxying for the regulatory, legal and political environment. These factors have proved to serve both as determinants in the M&A choices, as well as sources of performance explanation. Since they are of a primary focus in this paper, I elaborate on them further in the following sub-chapters.

2.2.3 Regulatory, legal and political factors in the context of M&A

Regulatory, legal and political factors and their relation to especially the cross-border M&A activities has become of particular interest for scholars. Increased activity as a result of regulation relaxation has been described by DeLong et al. (2001). Rossi & Volpin (2004) show that higher confidence of investors' leads to an increased level of transactions in the countries with better investors' protection and higher accounting standards. Lower shareholder protection comparing has been found category especially in case of the French law origin in the fundamental study of rule of law by La Porta et al. (1998) which distinguishes common vs. civil law-based countries. However, the stability of the relation of law origin and investors' protection was disputed by Martynova & Renneboog (2008), who develop corporate governance index incorporating the shareholders', minority shareholders' and creditors'

protection strength. Subsequently, Moeller et al. (2016) provide a comprehensive study of M&A market attractiveness in relation to regulatory, economic, financial, technological, socio-cultural and political maturity in the countries engaging in transactions and broaden the field of previously considered variables. Following the logic of their factor classification, a research overview of M&A performance with respect to regulatory and legal (with a specific focus on investor's protection), as well as political aspects is provided.

Regulatory and legal aspects comprise both the general factors describing the overall environment in the country such as regulatory quality, ease of starting a business, rule of law or contract enforcement (Moeller et al., 2016), as well as the corporate governance aspects. Investors' protection level has been especially of interest in the academic field. Evidence of positive effects on abnormal returns stemming from legal systems ensuring higher protection of shareholders is provided e.g. by Moeller & Schlingemann (2007). Yet, regulatory quality itself was found to be contributing positively to the M&A activity (Dodge & Barkdulll, 2009), too. Corporate governance index by Martynova & Renneboog (2008) accounts also for the interaction of the investors' protection metrics with the rule of law, as La Porta et al (1998) assume that stronger law enforcement may compensate for the lower protection. This hypothesis is also confirmed by the results of Kuipers et al. (2009).

Furthermore, cross-border M&A provide both target and acquiring companies with opportunities for improvement of corporate governance standards. Martynova & Renneboog (2008) describe theories of positive spillover by law (target's acquisition of higher standards from the acquirer thanks to the change in entity nationality), negative spillover by law (target's decrease in standards as a result of lower acquirer's lower standards level) and the bootstrapping hypothesis (acquirer's voluntary takeover of higher standards from the target). Their results support the positive spillover effect and bootstrapping hypothesis. These findings along with the higher frequency of acquisitions with low protection level targets by acquirers with high protection level are confirmed by Goergen & Renneboog (2008). Furthermore, Bris et al. (2008) show that the positive spillover by law contributes to increased Tobin's Q in the horizontal mergers through the adoption of improved shareholder's protection and accounting standards level.

The effects of corporate governance standards may also differ based on deal-specific characteristics. The acquirer's gain is higher with increasing corporate governance level only in the case of stock deals (Starks & Wei, 2004). With increasing investors' protection, public target companies are associated with negative returns, whereas the private targets yield

significantly positive returns for the acquirer (John et al. 2010). This might be explained by the positive correlation of the superior level of corporate governance with the amount of premium paid in case of a public target.

In terms of political factors, Moeller et al. (2016) include corruption perception, control of corruption, sovereign debt rating and political stability (as defined by Kaufman et al., 2010) as proxies for the level of country's maturity in this aspect. Dodge & Barkdull (2009) also add voice and accountability variable and confirm its positive effect on M&A activity by the analysis. Other studies suggest that political instability may stem from national elections (Julio & Yook, 2012) or prevalence of the political opinions within various political bodies (Yantek & Gartell, 1987). Moeller et al. (2016) interestingly find that M&A activity increases with political instability and decreasing maturity. They suggest that the inverse relationship can be explained by the tendency to exploit the perceived opportunities in the market despite the increased risk and is not only found in empirical research but also propounded by the M&A industry professionals. However, Dodge & Barkdull (2009) assume political stability and control of corruption (as defined by Kaufmann et al. (2010)), to be positively associated with the level of foreign direct investments in the host country due to the unpredictability of the environment. They admit, however, that the presence of corruption in the economy may contribute to a faster pace of business development by "greasing the wheels" (Dodge & Barkdull: p.9). The positive relation between the increased level control for corruption resulting in higher returns for the bidder in the M&A process was confirmed later by Ellis et al. (2011).

2.3 Hypotheses Development

Based on the theory outlined above and the industry-specific features that occur in practice, I decide to investigate the previously not vastly researched area and factors that may have implications for the parties involved in the M&A process. On these grounds, I develop hypotheses with respect to the (1) regulatory factors, (2) political factors in case of M&A; (3) differences in performance if private vs. public company is acquired; within the cross-border M&A area then also (4) difference in regulatory factors' levels, (5) difference in investor's protection; and (6) difference in the political factors' levels.

The regulatory and political environment represents an uncertainty upon undertaking an acquisition, given the regulated nature of the industry. Since this feature reduces the potential value creation for the acquiring shareholder and affects the M&A process, a well-developed environment is desirable and should induce a positive reaction in the stock performance upon the announcement. Although the higher level of these aspects could induce a higher premium

paid, leading to lower value creation, as the industry deals generally concern private companies, the liquidity discount is, therefore, common and reduces the probability that the former occurs. Considering the previous empirical results (e.g. Berry et al., 2000), we suggest the following hypotheses:

Hypothesis 1: There is a positive causal relationship between the level of regulatory factors in the target's country and the stock performance of the acquirer upon the acquisition announcement.

Hypothesis 2: There is a positive causal relationship between the level of political factors in the target's country and the stock performance of the acquirer upon the acquisition announcement.

Public target deals suffer comparing to private target ones from lower return for the acquirer due to the premium paid upon acquisition (Moeller et al., 2004). Countries with lower protection may lead to a lower value of the acquired companies due to the potentially higher agency costs. Therefore, the premium paid is lower and the acquirer benefits more from the acquisition (John et al., 2010). Furthermore, John et al. (2010) suggest that the positive reaction upon M&A announcement is lower in the case of private targets in the countries with lower investor protection level. As the target is acquired by a company from a higher-protection level country, the acquirer can benefit from a higher potential of agency costs reduction due to the higher insider ownership in the private companies. According to their premise, this factor plays a more important role than the potential illiquidity discount upon the acquisition, as they assume the private companies to be generally smaller. Yet, we suggest that given that the infrastructure companies are generally larger, the illiquidity discount is not outweighed by the agency cost effect. Therefore, in order to analyze potential differences in the investor's protection level given the public status of the target, we test the following hypothesis is tested for both public and private targets.

Hypothesis 3: There is a negative relationship between the level of investors' protection in the target country and the stock performance of the acquirer upon the acquisition announcement.

Acquirer's returns upon the cross-border M&A are generally found to be negative as they face often not a well-known environment. The importance of regulation and law practices as suggested e.g. by Kuipers et al. (2009) may, therefore, play an important role in an acquisition within a regulated industry. Target company in a country with a lower level of the regulation thus may induce a greater negative reaction in the stock market due to the aspects such as

potential uncertainty of the regulation with respect to the allowed rate or the approval of the M&A itself. Therefore, our next hypothesis is as follows:

Hypothesis 4: There is a negative causal relationship between the difference in regulatory factors level in the acquirer's and target's country and the stock performance of the acquirer upon the acquisition announcement.

Cross-border M&A may contribute to the adoption of better corporate governance leading to Pareto improvement (Bris et al., 2008) and the impact of varying levels of investor's protection on the M&A performance is positive due to the spillover by law effect (Martynova & Renneboog, 2008). Given that this legal aspect is related to the corporate governance rather than to the regulatory environment in the specific country which affects the infrastructure market potentially to a higher level, we investigate the investor's protection factor separately from the Hypothesis 4.

Hypothesis 5: There is a positive causal relationship between the difference in investor's protection level in the acquirer's and target's country and the stock performance of the acquirer upon the acquisition announcement.

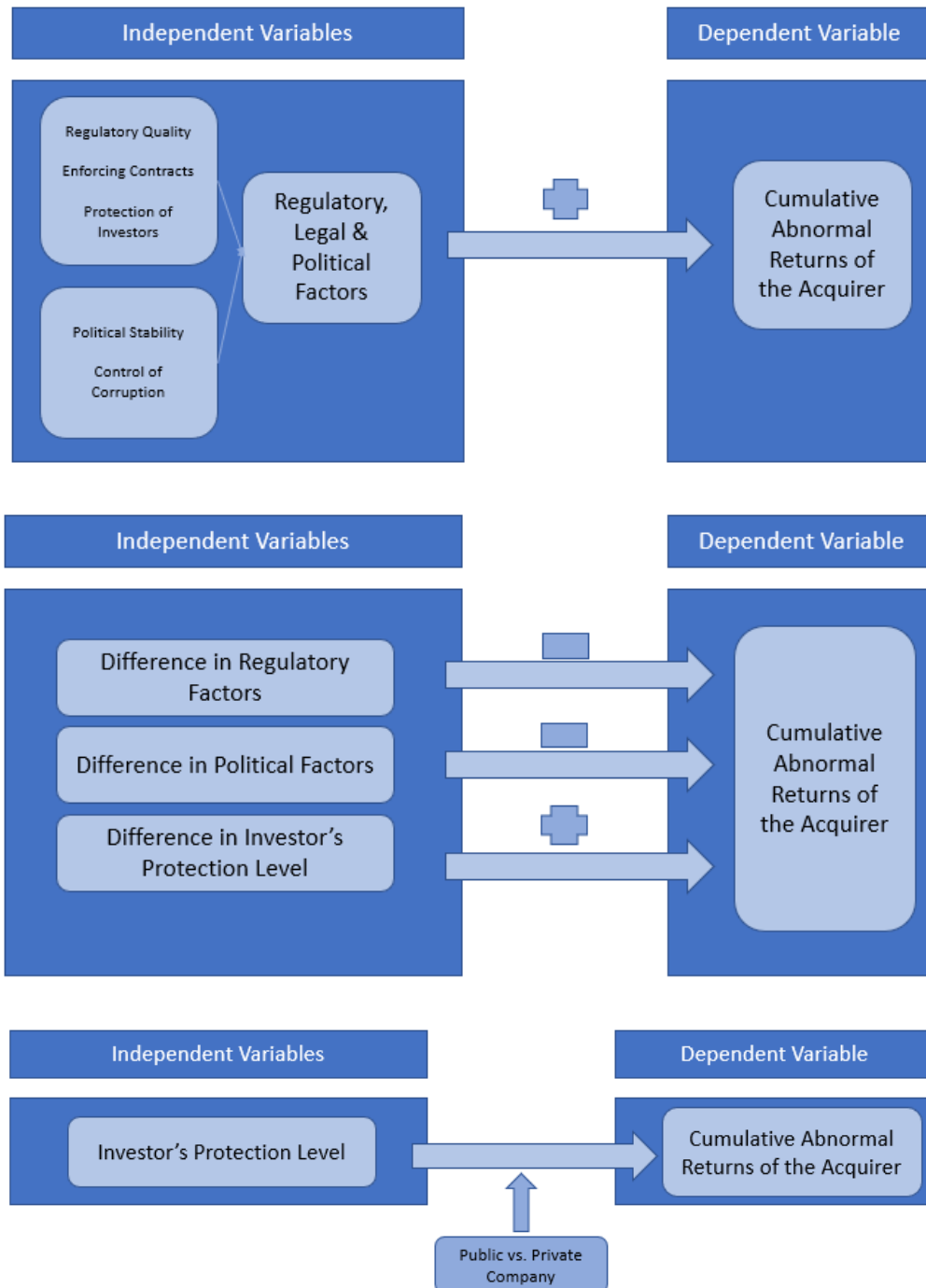
The political environment, maturity or stability is perceived as a core risk in the infrastructure sector (Altman, 2005). Therefore, on the grounds of previous research (e.g. Ellis et al., 2011) the stock market reaction to the acquisition of a company in the country with a lower political maturity level should be more negative with respect to the acquirer. Unlike the corporate governance factors described above, it is hard to imagine that the political environment would be subject to a spillover effect evidenced in cross-border M&As.

Hypothesis 6. There is a negative causal relationship between the difference in political factors level in the acquirer's and target's country and the stock performance of the acquirer upon the acquisition announcement.

The above-described hypotheses aim at bringing more insight into the explanatory power of these variables with respect to the cumulative abnormal returns realized by acquirers and targets upon acquisition announcements. Furthermore, their objective is to compare the coincidence of the results with the previously developed theories such as positive spillover by law (Martynova & Renneboog, 2008) or private vs. public company outcome differences (John et al., 2010).

Figure 1: Conceptual Framework

Figure 1 depicts the conceptual framework representing the base for hypotheses of this thesis that are subsequently tested. The framework is further divided into the sub-sections: (1) relationships between simple independent variables describing the regulatory, legal and political maturity in the relevant target country and the acquirer's CARs; (2) relationships between differences in the acquirer's and target's country-level variables and the acquirer's CARs; (3) relationship between target country investor's protection level and acquirer's CARs and condition of whether a private or a public company is acquired.



3. Methodology & Data

3.1 Methodology

Measurement of M&A performance has been a broadly discussed topic in academic research since 1960s. The plethora of methods, explanatory variables and multidisciplinary measures potentially causes inconsistency or incomparability in terms of performance evaluation given varying timeframes, objectiveness or level of analysis (Das & Kapil, 2012). As described in the chapter on M&A performance, results regarding the value-creating character of M&A vary greatly. Following the performance measure classification into financial performance variables, market performance variables, other non-financial operational ones and subjective ones presented by Das & Kapil (2012), a brief overview is included.

Accounting-based measures focus usually on variables describing growth or return. Although the interpretation of accounting ratios is straightforward and data generally easily retrievable, this measure of performance assessment suffers from several flaws. These include potential creative accounting and earnings management, changes in accounting policies or simple backward-looking nature of accounting measures. Furthermore, according to Bild et al. (2002), the measures traditionally used for M&A performance evaluation are not directly related to the NPV outcome, therefore, cannot be used to measure the M&A performance following the acquisition.

Market-based measures analyze either short-term or long-term effects of the acquisition through the event study methodology evaluating cumulative abnormal returns (CARs) or buy-and-hold returns (BHARs). The greatest strength of this evaluation measure is the forward-looking nature of the metric. As such, we measure the M&A performance by evaluation of the discounted present value of future cash flows. However, market-based measures also have several weaknesses, mainly stemming from the assumptions on which they are based such as market efficiency resulting in immediate pricing in of the announced event (Richard et al., 2009). Furthermore, the announcement is assumed to be unprecedented. Yet, potential information leakage is reflected in the stock prices already before the event window which might therefore systematically underestimate the effect of such an event. The event study methodology also assumes no major unanticipated event, that would affect the entire during the respective event window. Besides, a proper analysis should be based on well-developed capital markets which might pose a problem when employing this methodology for some emerging economies (Bodie et al., 2014). Long-term event studies might be also affected by

the survivorship bias or various other events that are priced into the market throughout the analyzed period.

Among other objective measures, Tobin's Q describing the replacement value of firm's asset is the variable often examined by researchers. Such a mixed measure provides the possibility of implementing some of the factors ignored by accounting-based or market-based evaluation in the analysis (Das & Kapil, 2012).

M&A performance can be also evaluated in a subjective manner, using conceptual modeling or managerial surveys. Such potentially biased measures, however, naturally suffer from low to no possibility of generalization of the results (Das & Kapil, 2012). Besides, researchers such as Zollo & Meier (2008) emphasized the correlation of subjective and objective measures.

Accounting-based and market-based measures are by far the most used assessments of M&A performance. After evaluation of advantages and disadvantages of the approaches described above, a short-term event study is chosen for our analysis as the most appropriate measure, mainly due to our preference for forward-looking measures.

3.1.1. Dependent variable

Cumulative abnormal returns (CARs) are a common measure of the acquisition impact. The effect of revelation bias whereby the bidder, knowing his true value is revealed along with the combined value upon the announcement (Bhagat et al.), is ruled out. Therefore, CARs can act as a dependent variable. For the purpose of investigation of the event announcement effects on the security returns through event study methodology, we want to identify potential abnormal returns with respect to the market returns, which can be described in the following way:

$$AR_{it} = R_{it} - E(R_{it}|X_t)$$

where,

- $AR_{i,t}$ is the abnormal return of the stock i at time t ,
- $R_{i,t}$ is the observed return of the stock i at time t ,
- $E(R_{it}|X_t)$ is the expected return of the stock i at time t .

Firstly, returns of the stocks and respective markets are computed. We choose to employ the logarithmic daily returns. As a standard finance research methodology, it ensures that results will not be distorted by potential extreme values and the returns distribution is more likely to be normal. Returns are, therefore, computed as follows:

$$R_{it} = \ln\left(\frac{P_{i,t+1}}{P_{i,t}}\right)$$

where,

- $P_{i,t}$ denotes the price of the stock i at time t ,
- $P_{i,t+1}$ denotes the price of the stock i at time $t+1$,
- $\ln$ is a natural logarithm.

Secondly, returns of the stock have to be regressed on the returns of the respective market within the estimation window in order to obtain the expected return values for the event window itself. Several methodologies are mentioned by academics: constant mean-return model, market-adjusted-return model, market model, as well as various economic and financial models employing e.g. three Fama-French factors and others (see e.g. Fama et al. (1969), McKinlay (1997)). Given the shortcomings of certain methodologies such as omission of the market dynamics (mean-return model case) or disregard of the correlation of the respective security with the market (market-adjusted-return model), we employ the market model in the analysis. In this manner, both the exogeneity of the market and endogeneity of the specific company's dynamics are included in the analyzes. Major country indices are used as the market benchmarks for the company's returns in order to maintain an accurate representation of the market environment. The model can be rewritten as a standard OLS regression in the following way:

$$R_{it} = \alpha_i + \beta_i \cdot R_{MT} + \varepsilon_{it}$$

where,

- $R_{i,t}$ is the observed return of the stock i at time t ,
- α_i denotes the intercept,
- β_i denotes the regression line coefficient,
- R_{MT} is the market M return at time T ,
- ε_{it} is the value of a residual.

Thirdly, the abnormal returns within the event window surrounding the announcement of the transaction are computed as follows:

$$AR_{it} = R_{it} - \hat{\alpha}_i - \hat{\beta}_i \cdot R_{MT}$$

where,

- $AR_{i,t}$ is the abnormal return of the stock i at time t ,
- $R_{i,t}$ is the observed return of the stock i at time t ,

- $\hat{\alpha}_i$ and $\hat{\beta}_i$ are the estimates of intercept and regression line coefficient estimated in the estimation window,
- R_{MT} is the return of market M, at time T.

Since the event window for which we analyze the announcement effects consist of multiple days, cumulative abnormal returns in the event window starting at time t_1 and ending at time t_2 are computed as follows:

$$CAR_{(t_1, t_2)} = \sum_{t_1}^{t_2} AR_{it}$$

These are divided by the number of days in the respective event window and cumulative average abnormal returns (CAARs) are obtained. Furthermore, significance of the results is tested. This enables us to analyze the actual effect of the announcement and understand, whether the event induces returns that are significantly different from zero. This hypothesis is tested using a standard t-static test:

$$t_{\overline{CAR}} = \frac{\overline{CAR}_{(t_1, t_2)}}{\sqrt{Var(\overline{CAR}_{(t_1, t_2)})}}$$

Opinions on what estimation window and even window length is appropriate widely differ. Researchers previously employed estimation windows ranging from 50 days to 500 days, research conducted specifically on power and utilities sector used 100-day long estimation windows with 30 days gap (Berry, 2000; Bartunek et al., 1993) or 249-day long window without a gap (Datta et al., 2013). In terms of the event window lengths, two aspects have to be reflected in the analysis. Firstly, the necessity to separate the event in question (M&A announcement) from other market events requires using a short event window. Secondly, longer event windows allow for capturing the information leakage prior to the announcement itself which should be acknowledged in the analysis according to MacKinlay (1997). Therefore, it is necessary to find a balance between these two approaches in order to ensure representativeness and quality of the results. For the robustness check purposes, five event windows are employed in our analysis: primary event windows surrounding the announcement: [-5, 5], [-3, 3], [-1, 1] and event windows capturing the possible runup due to information prior to the announcement or insider trading: [-5, 0], [-3, 0]. Since our analysis assumes at least semi-strong form of market efficiency, we do not include a gap between the estimation and event window. Considering the differing approaches used in empirical studies, the respective

estimation windows for our event window abnormal return analysis are: [-155, -6], [-153, -4], [-151, -2], [-155, -6], [-153, -4], in other words, with 150-day long estimation windows.

3.1.2. Independent variables

Independent variables used to measure the effects on CARs is a set of country-level indices describing the regulatory, legal and political environment in a specific country. Focus on countries rather than firms themselves is preferred in this study in order to capture accurately the regulatory background. This would be hardly possible on the company's level, given the heterogeneity of the countries the companies are based in (Martynova & Renneboog, 2008). Furthermore, specifically in case of corporate governance correlation between country and firm-level indices has been shown by researchers. Countries are the primary determinants of the cost of good governance or regulation bonding (see Doidge et al., 2007).

Independent variables we use for our analysis can be divided into two categories: simple indices and index differences. As described in the section 3.1.4 Regression models, Hypotheses 1 and 2 are associated with the simple indices given that we focus on the overall level of the metric in the country. Furthermore, the interaction term of regulatory quality and investors protection is added to measure the possible compensation of lower level of in one of the metrics by the other one based on previous research by La Porta et al. (1998). Hypothesis 3 is tested through splitting sample based on targets coming from countries with higher or lower investors' protection comparing to the mean and subsequent CARs analysis. Hypotheses 4, 5 and 6 follow the methodology of differences used e.g. by Martynova & Renneboog (2008) capturing the inequalities between the countries engaging in the deals and (in case of regulations) potential for improvement that would occur in case the standards of the opposite deal party were adopted. The independent and control variables were checked for high levels of correlation, in other words multicollinearity. Subsequently, two variables (rule of law and corruption of officials) that were initially intended to be included, were discarded. Correlation matrices can be found in the Appendix D. Tables 1 and 2 present an overview of the variables including their expected effect on CARs.

3.1.3 Control variables

In addition to the standard independent variables described in the previous section, ten control variables are selected in order to capture deal-specific, acquirer's or target's specific or country-specific effects. Also, fixed effects in terms of time (day announced) are considered.

As the deal-specific factors, we control for the diversifying nature of the transaction and distinction between a cross-border or domestic transaction. Since for instance, diversifying deals usually do not yield higher returns (Berger and Ofek, 1995), we want to control for these effects. Acquirer's size serves as a managerial hubris proxy (Roll, 1986) possibly causing an overpayment in the deal (Moeller et al., 2004). As a proxy for risk, we also include the control variable of acquirer's leverage. Given that the sample includes also regulated utility sector targets (for which the upside is limited and therefore does not constitute actual market for itself), we control for this type of company through a dummy variable. Since our sample is worldwide, therefore including financial markets with different levels of development and sophistication, we control for market liquidity through turnover ratio of the respective stock. Last but not least, we control for the GDP growth in the target country and the cultural distance between the countries involved in the transaction measured by the Hofstede's distance index score in the target country. Full overview of the control variables selected for this study and their expected effects can be found in the Appendix B.

Table 1: Independent Variables Overview – Hypotheses 1, 2

Table 1 summarizes the notations, definitions and sources of the independent variables used in the models for testing Hypotheses 1 and 2. Furthermore, the expected effects in terms of coefficient sign for the effect on acquirer's CARs is included.

Denotation	Name	Description	Expected sign - acquirer
Panel A: Regulatory factors			
RQ	Regulatory Quality	The variable describes the perceptions of the government's ability to formulate and implement sound policies and regulations permitting and promoting private sector development as defined by Kaufmann et al. (2010). Index values based on World Bank's Governance Matters - Worldwide Governance Indicators.	Positive
EC	Enforcing Contracts	Time and cost to resolve a commercial dispute and the quality of judicial processes for men and women as defined by World Bank (International Bank for Reconstruction and Development, 2019). Index values based on World Bank's Doing Business Indicators.	Positive
PI	Protecting Investors	Minority shareholder's rights in related-party transactions and in corporate governance as defined by World Bank (International Bank for Reconstruction and Development, 2019). Index values based on World Bank's Doing Business Indicators.	Positive
Panel B: Political factors			
PS	Political Stability	The variable describes the perceptions of likelihood that the government will be destabilized or overthrown by unconstitutional or violent means or politically-motivated violence and terrorism as defined by Kaufmann et al. (2010). Index values based on World Bank's Governance Matters - Worldwide Governance Indicators.	Positive
CC	Control of Corruption	The variable describes the perceptions of the extent to which public power is exercised for private gain, accounting for both petty and grand forms of corruption and "capture" of the state by elites and private interests as defined by Kaufmann et al. (2010). Index values based on World Bank's Governance Matters - Worldwide Governance Indicators.	Positive

Table 2: Independent Variables Overview – Hypotheses 4, 5 and 6

Table 2 summarizes the notations, definitions and the description of the independent variables used in the models for testing Hypotheses 4, 5 and 6. Furthermore, the expected effects in terms of coefficient sign for effect on the acquirer's CARs is included.

Denotation	Name	Definition	Description	Expected sign - acquirer
Panel A: Regulatory factors				
ΔRQ	Difference in Regulatory Quality	$= RQ_a - RQ_t$	difference of the acquirer's country index value and target's country index value, based on indices described in the table 1	Negative
ΔEC	Difference in Enforcing Contracts	$= EC_a - EC_t$		Negative
ΔPI	Difference in Protecting Investors	$= PI_a - PI_t$		Positive
Panel B: Political factors				
ΔPS	Difference in Political Stability	$= PS_a - PS_t$		Negative
ΔCC	Difference in Control of Corruption	$= CC_a - CC_t$		Negative

3.1.4 Regression models

In order to analyze the magnitude of impact of our independent variables on the dependent ones, we use the multiple OLS regression. The hypothesis and respective equations describing them (in case of 3-day event window only for the sake of length) are described in this section.

The following regression aims at measuring the impact of regulatory aspects in the target country on the cumulative abnormal returns of the acquirer. It is therefore related to the Hypotheses 1:

$$CAR_{i,[-1,+1]} = \alpha + \beta_1 RQ_i + \beta_2 EC_i + \beta_3 PI_i + \beta_4 RQ * PI_i + \beta_5 DIV_i + \beta_6 CB_i + \beta_7 AS_i + \beta_8 STAT_i + \beta_9 RU_i + \beta_{10} AL_i + \beta_{11} GDP_G_i + \beta_{12} HI_i + \beta_{13} ML_i + \beta_{14} TimeDummy_i + \varepsilon_i.$$

Hypotheses 2 investigating closer the impact of political factors' level in the target country on the acquirer's and target's cumulative abnormal returns is described as follows:

$$CAR_{i,[-1,+1]} = \alpha + \beta_1 PS_i + \beta_2 CC_i + \beta_3 DIV_i + \beta_4 CB_i + \beta_5 DS_i + \beta_6 AS_i + \beta_7 STAT_i + \beta_8 RU_i + \beta_9 AL_i + \beta_{10} GDP_G_i + \beta_{11} HI_i + \beta_{12} ML_i + \beta_{13} TimeDummy_i + \varepsilon_i.$$

Furthermore, we want to look more into the effects of differences in the regulatory factors level in the acquirer's and the target's country in case of cross-border deals, as well as the impact of target company status (public vs. private) on the cumulative abnormal returns as described in Hypotheses 4 and 5:

$$CAR_{i,[-1,+1]} = \alpha + \beta_1 \Delta RQ_i + \beta_2 \Delta EC_i + \beta_3 \Delta PI_i + \beta_4 DIV_i + \beta_5 AS_i + \beta_6 STAT_i + \beta_7 RU_i + \beta_8 AL_i + \beta_9 GDP_G_i + \beta_{10} HI_i + \beta_{11} ML_i + \beta_{12} TimeDummy_i + \varepsilon_i.$$

Last but not least, we investigate the effect of discrepancy in political factors in the target and acquirer's country in the cross-border transactions as suggested by Hypothesis 6:

$$CAR_{i,[-1,+1]} = \alpha + \beta_1\Delta PS_i + \beta_2\Delta CC_i + \beta_3DIV_i + \beta_4DS_i + \beta_5AS_i + \beta_6STAT_i + \beta_7RU_i + \beta_8AL_i \\ + \beta_9GDP_G_i + \beta_{10}HII_i + \beta_{11}ML_i + \beta_{12}TimeDummy_i + \varepsilon_i.$$

3.2 Data

3.2.1 Data collection and pre-processing

There are several sources for the M&A data: (1) Thomson One database aggregating quantitative and qualitative information regarding M&A deals, (2) Mergerstat database including deals with at least one party being US-based company and (3) the Zephyr database covering worldwide transactions. Considering the fact that my thesis is not geographically-focused and the extent of data availability and coverage of the respective databases, I chose to retrieve data on M&A transactions from the Thompson One database. From the initial overall sample (criterion 1) of mergers and acquisitions with announced day between January 1, 2006 and December 31, 2017, we scoped down the number of deals by following criteria: (2) the target company belongs to the infrastructure sector (according to SIC industry codes; see Appendix A for further description); (3) current deal status is “completed”; (4) acquirer is a listed company; (5) change of control occurred during the transaction (the acquirer owns more than 50% shares following to the transaction which was not the case prior transaction) . Following this selection, our preliminary sample included 2, 233 M&A deals.

Furthermore, companies' stock prices and index levels were downloaded from Datastream financial database in order to compute returns and cumulative abnormal returns upon the even announcement. Given the number of countries from which public acquirers and targets come from, 45 indices were downloaded to provide the respective regional market benchmark. The independent variables were retrieved through World Bank Databank. As such, it provided two databases, World Governance Indicators and Doing Business Indicators, that were sources of our Regulatory Quality, Control of Corruption, Political Stability, Investors' Protection and Enforcing Contracts variables. Although Doing Business Indicators methodology was revised in 2016, according to the World Bank, comparing the results of 2016 given both the previous, as well as the updated methodology resulted in results and ranking with “*correlation coefficient very close to 1*” (The World Bank Group 2019). Therefore, we included the data with the assumption that the unsubstantial change ensures the indices are comparable across time.

Majority of the control variables was constructed based on the detailed data provided by the Thompson One M&A database. This was the case of: diversifying nature of the deal, regional dummy, total acquirer's assets, target company status and time fixed effects. World Bank Database provided the data on historical GDP growth rates for the countries in question. Market liquidity and total debt value necessary for the leverage computation were retrieved through Datastream.

Due to the availability of the data of stock prices, independent and control variables, our dataset was scaled down. Furthermore, we checked for the confounding events in the sample: we exclude the deals that happen in the same market within the respective event window, as well as deals made by the same public acquirer at the same time, as the effects on stock price would not be clear to disentangle. The issue of confounding events could be extended to the level of other events related to the acquiring company (e.g. earnings announcements), yet due to the limited data availability with regards to such events, I decided not to account for those. Moreover, the observations with the leverage values lower than 0 and greater than 1 were excluded due to possible data mistakes or inclusion of distressed or unusually operating companies. For the market liquidity variable measured as the average turnover ratio in the event window days, we excluded the observations with the value of 0, as it suggests that shares of such companies are not traded at all, which is not suitable for the analysis we want to perform. We also considered the fact that companies operating in gas distribution and telecommunications operate on a multinational level and the respective nation provided by the Thompson One database is based on the company's headquarters. For the companies in question, we aimed at adjusting the respective nation based on the top revenue country, as it is plausibly the one of which the regulation it must abide by. Yet, such revenue breakdown was available only for a fraction of deals, therefore, initial nationalities were kept. Overall, majority of the deal exclusions were due to the confounding events and not data availability.

Following this procedure, our final sample included 1106 observations for the regressions with simple index values and 310 observations in the subsample of cross-border deals for the regressions with index differences. Table 3 summarizes the deal selection process.

Table 3: Deals Sample Selection Criteria

Table 3 summarizes the selection criteria for the data sample used for the analysis. Data in the steps 1. to 5. Were retrieved in the Thompson One deals database.

Data sample - deals selection			
No.	Identifier	Decription	Deal count
1.	Time Period	Announced between January 1, 2006 and December 31, 2017	n/a
2.	Industry	Target falls within the category of infrastructure based on the primary SIC code (see Appendix A for a detailed description of selected categories)	31,352
3.	Deal Status	Completed or Unconditional	22,864
4.	Acquirer's status	Acquirer is a public company	3,602
5.	Stake acquires	Change of control in the transaction	2,233
6.	Data availability	Stock data, independent and control variables availability, data processing	1,106

3.2.2 Dataset descriptive statistics

The following sub-chapter provides an overview of the descriptive statistics of the dependent variables, cumulative abnormal returns, as well as the independent and control variables for the entire sample, as well as the cross-border deals subsample.

Cumulative Abnormal Returns descriptive statistics are summarized in the Table 4. For all 5 event windows considered for the sample size of 1433, the mean cumulative abnormal return is positive within the range from 0.5% to 1% suggesting small gains for the acquirers. We observe certain outliers according to the minimum and maximum values especially for the [-5, 5] window. Yet, the skewness of the distribution is low (within two-digit range used as a rule of thumb). As a no surprise for financial data, cumulative abnormal returns variables exhibit leptokurtic distribution suggested by high kurtosis scores. In terms of statistical significance, only the cumulative abnormal returns in the 3-day [-1, 1] and 4-day [-3, 0] windows are significant at 1% level. The event window [-5, 0] shows cumulative abnormal returns significant at 5% level, suggesting possible information leakage effects in the market prior to the actual announcement. This differed for the cross-border subsample, for which we identified two event windows investigated (specifically [-5, 0], [-3, 0]) significant at 10% and 5% level, respectively. Yet, the average cumulative abnormal return values were positive in all cases, within the 0.5 to 1% range for all event windows apart from the [-5, 5] one.

Descriptive statistics of the independent and control variables for the full sample are presented in the Table 5. In terms of independent variables, the means of the simple index variables are closer to the maximum rather than the minimum values for all of them. This suggests rather higher level of regulatory and political factor levels in the acquiring countries. In terms of the independent variables defined as differences between the acquirer's and target country, the

means are positive apart from the Investors' Protection variable. This suggests that the acquirer countries have rather higher levels of regulatory and political factors or are possibly more developed in these aspects. This is also in line with the potential spillover effect that might be induced by cross-border transactions. The negative sign of the mean in the Investors' Protection variable suggests acquirers might tend to pursue targets with better investor's protection. This is also commonly commented on in the literature (e.g. Martynova & Renneboog, 2008) as being a possible cause of either negative spillover effect (decrease in the corporate governance level post-transaction) or bootstrapping hypothesis (acquirer's voluntary deciding for higher corporate governance level).

In terms of the control variables, diversifying deals happened to form half of the dataset as the mean value is 0.5. Therefore, we also carried out subsequently also a subsample analysis to account for the difference arising from such a deal and compare the results. Majority of the deals in our dataset are domestic. This, therefore, slightly limits the subsample used to inspect the effects of differences between the country-level regulatory and political factors. Majority of the acquired companies were private and little more than a half of them belonged to the regulated utility sector. Acquirers exhibit rather higher leverage with the mean value of 55%. The average GDP growth value in the target countries is 2.9% which appears to be in line with the fact that majority of the years we investigate belong to the non-crisis period of economic development in the world. The outliers that occur in this variable were detected to be reasonable as they either characterize a substantial economic downturn (approximately -14.7% in case of Ireland during the crisis starting 2008) or boom (25.5% in case of the Vietnamese economy in 2015). As they represent the real economic development in the countries, they are kept in the sample. In terms of market liquidity, for all event windows considered, approximately 0.7% of shares outstanding are traded on an average.

The descriptive statistics for the cross-border subsample used for the regressions with differences in the regulatory and political factors' levels in the acquirer and target countries were found highly comparable with the subsample with respect to the decomposition of the dummy groups, as well as the mean values for the continuous variables. For this reason, they are reported in the Appendix C.

Table 4: Cumulative Abnormal Returns Descriptive Statistics

Table 4 presents descriptive statistics overview of the dependent variable, cumulative abnormal returns, for the full sample used for regressions with simple indices as independent variables and for the cross-border transactions subsample used for the regressions with differences in index levels as independent variables. CARs are based on the market methodology with the estimation period of 150 days prior to the event window. *, **, *** represent significance at 10%, 5% and 1% level, respectively.

Descriptive Statistics of Cumulative Abnormal Returns								
Variable	Obs	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	Significance
CARs (-5, 5)	1,106	0.0082	0.1449	-1.3987	1.4658	1.1141	29.1824	1.09
CARs (-3, 3)	1,106	0.0087	0.1202	-0.8117	1.0323	2.4285	25.1878	1.48
CARs (-1 1)	1,106	0.0095	0.0990	-0.4869	1.4475	6.0546	76.4385	***2.48
CARs (-5, 0)	1,106	0.0119	0.1185	-0.9160	1.5510	3.4024	45.5472	**2.15
CARs (-3, 0)	1,106	0.0105	0.1024	-0.5768	1.0647	4.0521	4.0521	***2.35

Descriptive Statistics of Cumulative Abnormal Returns - Crossborder Subsample								
Variable	Obs	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	Significance
CARs (-5, 5)	310	0.0024	0.1093	-0.7548	0.3680	-1.0170	11.9025	0.32
CARs (-3, 3)	310	0.0048	0.0897	-0.4037	0.5486	0.4971	10.2349	0.79
CARs (-1 1)	310	0.0047	0.0629	-0.3218	0.2749	0.1681	8.7473	1.18
CARs (-5, 0)	310	0.0103	0.1006	-0.5371	0.6735	1.3376	17.3124	*1.85
CARs (-3, 0)	310	0.0090	0.0870	-0.3932	0.6186	2.0963	19.3606	**1.96

Table 5: Independent and Control Variables Descriptive Statistics – Full Sample

Table 5 presents descriptive statistics overview of the independent and control variables, for the full sample used for regressions with simple indices as independent variables.

Descriptive Statistics of Independent and Control Variables								
Variable		Obs	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Independent variables								
PI	Investor's Protection	1,106	64.5165	14.6782	26.6700	96.6700	-0.1150	2.4454
EC	Enforcing contracts	1,106	66.4636	11.4545	29.0400	93.3600	-1.3664	5.4585
RQ	Regulatory Quality	1,106	1.0274	0.7945	-0.7375	2.2605	-0.7568	2.0878
CC	Control of Corruption	1,106	0.9432	1.0031	-1.1320	2.4700	-0.5608	1.9608
PS	Political Stability	1,106	0.3047	0.7232	-2.0091	1.5935	-0.8065	2.7240
Deal-specific control variables								
DIV	Diversifying deal	1,106	0.5009	0.5002	0.0000	1.0000	-0.0036	1.0000
CB	Cross-border deal	1,106	0.7179	0.4502	0.0000	1.0000	-0.9684	1.9378
Acquirer or target-specific control variables								
ln(AS)	Acquirer's size	1,106	6.3010	2.8779	-5.8091	13.6516	-0.3274	3.2455
STAT	Target's status	1,106	0.8553	0.3519	0.0000	1.0000	-2.0203	5.0816
RU	Regulated Utilities	1,106	0.5561	0.4971	0.0000	1.0000	-0.2257	1.0509
AL	Acquirer's leverage	1,106	0.5432	0.2043	0.0123	0.9968	-0.2669	2.6456
Country-specific control variables								
GDP_G	GDP Growth in the target country	1,106	2.8537	3.2599	-9.1325	25.5573	0.4336	6.9872
HI	Hofstede's Cultural Index	1,106	52.6528	19.2440	11.0000	95.0000	0.5781	2.2427
ML (-5, 5)	Market liquidity in the event window (-5, 5)	1,106	0.0069	0.0167	0.0000	0.2890	9.8805	140.0633
ML (-3, 3)	Market liquidity in the event window (-3, 3)	1,106	0.0074	0.0192	0.0000	0.3028	10.4192	147.5330
ML (-1 1)	Market liquidity in the event window (-1, 1)	1,106	0.0080	0.0212	0.0000	0.3464	10.3140	147.0126
ML (-5, 0)	Market liquidity in the event window (-5, 0)	1,106	0.0067	0.0179	0.0000	0.3206	10.9820	170.0082
ML (-3, 0)	Market liquidity in the event window (-3, 0)	1,106	0.0071	0.0208	0.0000	0.3751	11.7996	184.4987

4. Results and Discussion

In the following section, we present the findings of our analysis for the 7-day long event window $[-3, 3]$ around the announcement day. First, we provide comments on testing and solutions regarding the satisfaction of the crucial OLS assumptions. Second, we present and interpret the actual results for the respective hypotheses of this thesis. Third, a brief discussion of the findings regarding the control variables is presented. Finally, results and discussion of the robustness checks close the chapter.

4.1 Testing for multicollinearity, heteroskedasticity and autocorrelation

Prior to the actual regression analysis, the assumptions of OLS models must be checked. In our case, we considered multicollinearity, heteroskedasticity and autocorrelation.

Presence of multicollinearity would cause overestimation of the goodness of fit of our model. Therefore, this condition was checked for the intended models through correlation matrices and VIF scores for the respective variables. Some of the variables exhibit moderate to high correlation, specifically Regulatory Quality and Hofstede's Cultural Index (-0.86), Control of Corruption with the Hofstede's Cultural Index and Political Stability (-0.85, in both cases) and differences in the Control of Corruption and the Political Stability (0.81). Yet, none of the variables has scored higher than the generally acceptable VIF score of 10. Also, apart from the simple indices of Control of Corruption vs. Political Stability and Regulatory Quality vs. Hofstede Index, the stricter VIF score limit of 5 was not breached. Full results of the multicollinearity testing are presented in the Appendix D.

All regression models were tested for the heteroskedasticity presence. Since it was detected in all cases at 1% significance levels through Wald test, heteroskedasticity adjusted standard errors had to be employed. Furthermore, autocorrelation was tested through Breusch-Godfrey and Durbin-Watson tests for the respective CARs regressions. The results of Durbin-Watson test suggest limited concern of the autocorrelation issue, as the scores were mostly within the range of 1.85 and 2.25. Breusch-Godfrey test would reject the null hypothesis of zero autocorrelation in majority of the CARs regressions at least at 10% level. Therefore, I decide to treat the autocorrelation issue. In the end, robust standard errors are employed in all regressions and the subsequent results are reported in this thesis.

4.2 Results Discussion

4.2.1 Hypothesis 1

Hypothesis 1 assumed a positive relationship between acquirer's CARs upon the M&A announcement and the level of regulatory factors in the target countries. Regressions run employed one index, combinations of two of them or all three of them and an interaction effect. Furthermore, separate subsamples of diversifying and within-industry deals were analyzed. Fixed effects models were generally preferred, as the null hypothesis of the Hausman test rejected the random effects models as the appropriate for a vast majority of models either at 5% or 10% significance level. Results of the models employing all regulatory factor indices and the interaction term are reported in the Table 6 and further comments can be found in the following paragraphs.

The variable that was found the most significant, specifically 1% level in the model C in the full sample, was the Regulatory Quality. Similarly to most of the coefficients of other regulatory factor indices, it showed a positive relationship between the level of the index and the CARs obtained upon the announcement. On the contrary, the Investor's Protection and the interaction term of Investors' Protection was found statistically significant at 10% level for the full sample models A and B with the implied effect on CARs to be negative. Looking at the specific components in case of 1 unit increase in the Investors' Protection index (*ceteris paribus*) the overall incremental effect on CARs would be negative between 0.07 and 0.14 percentage points, which does not seem to be an economically significant return increment. I assume that investors may overpay or pay higher premia when acquiring targets in countries with better investors' protection, leading to a lower or negative effect on their CARs. This, on the other hand might not be the case for the other indices included as independent variables. The Enforcement of Contracts variable was found to have a positive impact on the acquirer's CARs, yet the coefficients obtained were not statistically significant and the economic impact is insignificant, as well. Generally, the Regulatory Quality with the greatest coefficient and significance was identified as the factor having the greatest influence on the acquirer's CARs.

In case of the Regulatory Quality increase by 1 unit (*ceteris paribus*), the overall effect on CARs would stay positive of approximately 12.26 percentage points (according to the model C), which can be considered as substantial in terms of economic significance. Although the results suggest that certain regulatory aspects yield contradictory effects or might potentially offset each other thanks to the negative interaction term, Regulatory Quality, in this case, appears to represent a more fundamental factor on the target country environment leading to

positive effects on CARs. Given the negative sign of the interaction effect, the higher protection of investors apparently does not create value and does not compensate for lower regulatory quality. This is in line with previous findings of La Porta et al. (1998) or Kuipers et al. (2009) suggesting that lower investors' protection levels can be offset by stronger quality regulations or their enforcement. Since the most significant variables overall suggest positive effects on the acquirer's CARs, we conclude that the results support the Hypothesis 1. These results were also confirmed by mostly positive significant coefficients obtained from the regressions employing one or a combination of two indices.

A separate analysis of the subsample of diversifying and inter-industry transactions followed. Selection of significant results obtained is also reported in the Table 6. Comparing to the full sample, Enforcing Contracts variable was significant in many more regressions tested and the opposite was true for the Regulatory Quality variable. Furthermore, the results suggest that a higher level of Enforcing Contracts should have a positive effect on acquirer's CARs, which supports the Hypothesis 1. Moreover, our results support our hypothesis that Enforcing Contracts is a factor influencing CARs of the acquirer, which (to my knowledge) in the previous empirical research was not employed or tested. Diversifying deals subsample also exhibited significance of regulatory factors comparing to the within-industry ones. This might be explained by possibly lower insecurity regarding the regulatory environment in a different sector that the acquirer's which induces positive market reaction. On the contrary, this might not be of such importance in case on the inter-industry deals where the environment of the target company might be more familiar for the acquiring company. Although some differences in the subsamples results comparing to the full sample occurred, the significant results generally support them and thus corroborate the Hypothesis 1.

Our findings, therefore, support previous research (see e.g. Dodge & Barkdull, 2009 or Berry et al., 2000) suggesting positive effects of superior regulatory aspects in the target country on the M&A activity. I assume that despite previously discussed higher premium that might be paid in case of an acquisition in a target country more highly ranked in terms of investors' protection, the market still appreciates the lower uncertainty that this acquisition represents comparing to one with a target in a country with e.g. insufficient protection of the investors or lower enforcement possibilities in terms of contracts, subsequently leading even to potential unattainability of business goals. Thus, transactions increasing certainty regarding the environment directly impacting the operations of the acquired business are treated by the market as value-creating.

4.2.2 Hypothesis 2

In the Hypothesis 2, we assumed the positive relationship between acquirer's CARs upon the transaction announcement and the levels of the political factors in the target country. Testing of this hypothesis was carried out through both random effects model, for which the null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 10% level model A including only the Control of Corruption index. Despite random effects not being rejected for all other models, we decided to elaborate further on the time fixed effects regressions, as the adjusted R^2 appeared higher, thus we assume the model to be specified in a better way. These results can be found in the Table 7.

For the models tested, we found inconclusive results in terms of signs of the independent variables' coefficients. Control of Corruption variable exhibited a positive effect on the acquirer's CARs significant at 10% level for the full-sample regressions. The higher levels of Control of Corruption represent a more transparent environment, thus less uncertainty regarding the environment which is translated into the positive market reaction. On the contrary, the analysis relationship between Political Stability and CARs yielded mixed signs of the coefficients. However, since the coefficients of Political Stability variable were not found significant, this suggests that the prevalent effect is coming from the Control of Corruption and is positive. Since the assumed relationship between political factors level and the CARs was positive, the Hypothesis 2 can be supported by the significant results obtained from the models tested.

Further analysis was carried out on subsamples of diversifying and non-diversifying mergers due to the high presence of diversifying transactions in the sample. Although for the diversifying deals, the results were not found significant, highly comparable results were obtained for the subsample of transactions within the same industry. Control of Corruption was found significant at 10% level for the within-industry models. On the other hand, we never found Political Stability significant and even the signs of coefficients often changed. Little inference can be done, therefore, with respect to this variable. Since the results obtained for the non-diversifying deals subsample are in line with the full sample, this provides further support for them.

We observe that the uncertainty of the future value creation stemming from e.g. not certain or less controlled environment in the target country may induce negative market reaction. Potentially, the premium paid to targets is not positively correlated with the superiority of the political environment in the target country, as a higher level of the political factors employed

has a positive effect on the acquirer's CARs. Our results thus corroborate previous findings of e.g. Dodge & Barkdull (2009) or Ellis et al. (2015).

Table 6: Hypothesis 1 Results – Regulatory Factors Levels – Time Fixed Effects

Table 6 summarizes results of the Hypothesis 1 measuring the relationship of the regulatory factors' levels in the target country and CARs. Selection of regression results with significant independent variable coefficients is presented. CARs of the event window [-3,3] used as the dependent variable in the time fixed effects models. Respective t-statistics for are reported in brackets. The null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 5% level in case of the all models for the full sample and at 10% level in case of the models A and B with diversifying deals. Fixed effect models are primarily reported due to the better overall goodness of fit and for the sake of unification. Significance levels are estimated using the robust standard errors. *, **, *** represent significance at 10%, 5% and 1% level, respectively. Robust standard errors were employed to correct for autocorrelation and heteroskedasticity.

CARs [-3, 3] mean = 0.0085		Full Sample			Diversifying deals		
Variable		A	B	C	A	B	C
Independent variables							
PI	Investor's Protection	*-0.0012 (-1.67)	*-0.0014 (-1.78)	0.0012 (1.04)	***-0.0029 (-2.38)	** -0.0026 (-1.95)	-0.0018 (-0.57)
EC	Enforcing contracts	0.0010 (1.20)	0.0007 (0.68)	0.0011 (1.17)	***0.0028 (2.48)	***0.0031 (-2.30)	*0.0034 (1.94)
RQ	Regulatory Quality		0.0191 (0.78)	***0.1245 (2.78)		-0.0194 (-0.61)	0.0144 (0.15)
PI*RQ	Interaction of Investors' Protection and Regulatory Quality			***-0.0019 (-2.70)			-0.0006 (-0.32)
Deal-specific control variables							
DIV	Diversifying deal	***0.054 (3.1)	***0.0549 (3.15)	***0.0603 (3.46)			
CB	Cross-border deal	-0.0204 (-1.17)	-0.0194 (-1.11)	-0.0185 (-1.07)	0.0270 (0.78)	0.0283 (0.81)	0.0289 (0.81)
Acquirer or target-specific control variables							
ln(AS)	Acquirer's size	***-0.0139 (-2.73)	***-0.0138 (-2.69)	***-0.0141 (-2.78)	-0.0107 (-1.03)	-0.0109 (-1.06)	-0.0105 (-0.95)
STAT	Target's status	0.0138 (0.50)	0.0139 (0.51)	0.0107 (0.40)	*0.1287 (1.91)	*0.1282 (1.92)	*0.1255 (1.78)
RU	Regulated Utilities	-0.0013 (-0.08)	-0.0008 (-0.05)	-0.0040 (-0.26)	0.0449 (1.52)	0.0463 (1.58)	0.0437 (1.41)
AL	Acquirer's leverage	-0.0165 (-0.37)	-0.0199 (-0.43)	-0.0308 (-0.68)	0.0677 (0.91)	0.0677 (0.92)	0.0641 (0.90)
Country-specific control variables							
GDP_G	GDP Growth in the target country	*-0.0056 (-1.82)	*-0.0053 (-1.74)	*-0.005 (-1.66)	-0.0019 (-0.16)	-0.0039 (-0.30)	-0.0029 (-0.26)
HI	Hofstede's Cultural Index	-0.0002 (-0.33)	0.0004 (0.39)	0.0004 (0.41)	-0.0019 (-1.03)	-0.0023 (-1.46)	-0.0022 (-1.29)
ML (-3, 3)	Market liquidity	*1.0871 (1.89)	**1.1088 (1.99)	0.9717 (1.64)	***-3.8481 (-2.96)	***-4.0202 (-2.8)	***-4.1725 (-2.43)
	Constant	0.1088 (1.14)	0.0942 (1.01)	-0.0604 (-0.59)	0.0161 (0.10)	0.0267 (0.17)	-0.0363 (-0.12)
n		1106	1106	1106	552	552	552
Adjusted R2		13.49%	13.61%	14.91%	28.55%	28.64%	28.62%
F-statistic		1.59	1.51	1.81	2.97	2.94	2.83

Table 7: Hypothesis 2 Results – Political Factors Levels – Full and Within-industry Deals – Time Fixed Effects

Table 7 summarizes selected results of the Hypothesis 2 measuring the relationship of the levels of the political factors in the target country and CARs. CARs of the event window [-3, 3] used as the dependent variable in the time fixed effects models. Respective t-statistics are reported in brackets. The null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 10% level in case of the models A both in full and within-industry deals. For the other models, random effects were not rejected as inappropriate specification, yet, they are reported due to the higher R^2 delivering and for the sake of unification. Significance levels are estimated using the robust standard errors. *, **, *** represent significance at 10%, 5% and 1% level, respectively. Robust standard errors were employed to correct for autocorrelation and heteroskedasticity.

CARs [-3, 3] mean = 0.0085		Full sample			Within industry deals		
Variable		A	B	C	A	B	C
Independent variables							
CC	Control of Corruption	*0.0196 (1.34)		*0.0334 (1.68)	*0.0585 (1.75)		*0.0840 (1.73)
PS	Political Stability		0.0036 (0.21)	-0.0204 (-0.88)		0.0324 (0.65)	-0.0420 (-0.59)
Deal-specific control variables							
DIV	Diversifying deal	*0.0511 (3.03)	0.0496 (2.90)	***-0.0501 (2.93)			
CB	Cross-Border deal	*-0.0186 (-1.06)	-0.0182 (-1.03)	-0.0187 (-1.07)	-0.0668 (-1.50)	-0.0663 (-1.42)	-0.0598 (-1.27)
Acquirer or target-specific control variables							
ln(AS)	Acquirer's size	*-0.0129 (-2.55)	***-0.0095 (-2.63)	***-0.0125 (-2.47)	***-0.019 (-2.44)	***-0.0203 (-2.55)	-0.0181 (-0.79)
STAT	Target's status	0.0166 (0.63)	0.0136 (0.50)	0.0200 (0.74)	-0.0468 (-0.87)	-0.0518 (-0.95)	-0.0414 (-0.79)
RU	Regulated Utilities	-0.0006 (-0.04)	-0.0027 (-0.17)	0.0005 (0.03)	-0.0293 (-0.91)	-0.0350 (-1.11)	-0.0305 (-0.96)
AL	Acquirer's leverage	-0.0252 (-0.55)	-0.0244 (-0.53)	-0.0285 (-0.61)	-0.1071 (-1.21)	-0.0965 (-1.09)	-0.1280 (-1.46)
Country-specific control variables							
GDP_G	GDP Growth in the target country	*-0.0049 (-1.67)	*-0.0052 (-1.75)	-0.0051 (-1.74)	**_-0.0182 (-2.09)	-0.0175 (1.90)	*-0.0175 (-1.93)
HI	Hofstede's Cultural Index	0.0011 (1.18)	0.0003 (0.47)	0.0012 (1.28)	0.0034 (1.66)	0.0015 (0.87)	*0.0034 (1.68)
ML (-3, 3)	Market liquidity	**1.144 (2.03)	**1.1315 (1.97)	**1.1727 (2.04)	***1.4941 (5.02)	***1.442 (4.65)	***1.529 (5.00)
	Constant	0.0051 (0.07)	0.0726 (1.36)	-0.0100 (-0.14)	0.1074 (0.81)	0.2563 (1.92)	0.0900 (0.71)
n		1106	1106	1106	554	554	554
Adjusted R2		8.00%	12.89%	12.81%	32.90%	30.28%	33.44%
F-statistic		1.11	1.73	1.73	1.77	3.5	5.47

4.2.3 Hypothesis 3

Hypothesis 3 tested the relation of investors' protection level and cumulative abnormal returns upon the announcement with respect to the public status of the target company. Firstly, a public target situated in a country with a higher protection level should yield lower CARs for the acquirer. This is based on the premise that the premium offered in the acquisitions is related to the quality and riskiness of the companies depending on the level of this independent variable (John et al., 2010). Our results dispute this hypothesis, as the CARs of public targets in low protection level countries are lower than in higher ranked countries. From the economic standpoint, the difference between average CARs in higher vs. lower investors protection

country is quite significant (approximately 1.89 percentage points). Yet, the results obtained for the CARs of the 7-day event window of [-3, 3] were not statistically significant. The inference from these results is, therefore, limited. Since premium was assumed as the main driver behind the CARs level in the hypothesis, I conclude that the investors' protection level is not a highly substantial determinant of the premium paid upon the acquisition. I suggest, therefore, there might be other factors that can cause the differences within premia paid for public targets such as competition in the acquisition, acquirer's industry proximity and other. These potentially influence the acquirer's CARs upon acquisition on a more systematic level.

Furthermore, I hypothesize that private targets exhibit higher returns when located in countries with lower investors' protection levels. The results obtained from our analysis corroborate this hypothesis and the acquirer's CARs for private targets in the lower investor's protection level countries were found significant at 1% level. Furthermore, the result of 1.94% CAR in the 7-day long event window is considerably economically significant, as well. As such, we refute the hypothesis suggested previously by John et al. (2010). They suggested that for private targets, countries with higher investors' protection levels contribute to higher acquirer's CARs due to higher potential of agency costs reduction, as the insider ownership in private companies is higher. According to John et al. (2010) that this factor plays more importance than a liquidity discount (or lower investors' protection level discount). Since the assumption of the private companies in case of the paper of John et al. (2010) was that they are rather smaller, I argue that since infrastructure firms are usually rather large, the illiquidity discount is not offset. The discount placed on private companies due to illiquidity (and possible higher information asymmetry due to their absence at the market) is probably higher in the target countries in the lower section of investors' protection ranking, as the uncertainty level is increased and is even magnified by the size of companies within the analyzed sector.

Table 8: Hypothesis 3 Results – Investors' Protection Level Relationship with Cumulative Abnormal Returns for Private and Public Companies

Table 8 presents results for the Hypothesis 3 testing the relationship between level of Investors' Protection level in the target country and cumulative abnormal returns with respect to the subgroups of public vs. private targets. Results are reported for the analysis carried out on the CARs of 7-day event window [-3, 3] and the full sample of 1106 transactions.

	CARs [-3, 3] average	
Subgroup	Public companies	Private Companies
Higher Investor Protection Countries	0.82%	-0.17%
	0.90	-0.12
Lower Investor Protection Countries	-1.07%	1.94%
	-1.24	***4.21

4.2.4 Hypothesis 4

Hypothesis 4 assumed a negative causal relationship between an increasing difference in the regulatory factor's level between the acquirer's and target's country and the acquirer's CARs increases. This might be induced e.g. by the uncertainty regarding the regulations in the respective target country. We performed both random and time fixed effects regressions. The null hypothesis of the random effects model being the appropriate specification was rejected by the Hausman test at 5% significance level for the models A, B, C, D, E and G; and at 10% level for the model F. These results are also presented in the Table 9. Regarding the respective indices used for measuring the differences in the regulatory factor levels, Investors' Protection level difference was found significant at least at 5% level in all models. Since there was a separate hypothesis regarding this aspect, it is further commented in the section 4.2.4. Furthermore, Difference in Regulatory Quality and Enforcing Contracts were not significant in the models reported.

Difference in Enforcing Contracts did not support the Hypothesis 4 by the signs in any of the models tested, yet, it was not found significant. Difference in Regulatory Quality did support the Hypothesis 4 in model D and G, yet the coefficients were not statistically significant either. Overall, the results suggest that the effect of differences in the regulatory factors levels on acquirer's CARs is positive. However, the economic significance of the results is quite low, apart from the Regulatory Quality variable when included in combination with another index as independent variables. Since we assumed the effect of differences to be negative, the actual Hypothesis 4 is not supported by the results obtained. Previous literature (Rossi & Volpin, 2004) suggested the regulatory environment as an important driver for undertaking the M&A itself. The significance of this finding was, however, disputed by Moeller et al. (2016), as no evidence supported the determination of M&A by the regulatory environment. Our results add to the lack of evidence, as also the effects on acquirer's CARs are insignificant. Yet with the positive effect found in the results. Apparently, the differences in the regulatory factor levels are rather perceived as opportunities for instance for a better corporate governance adoption as suggested by Bris et al. (2008) and Martynova & Renneboog (2008).

Multicollinearity was detected in case of several variables in the subsamples of diversifying and non-diversifying mergers, therefore a separate analysis of diversifying and non-diversifying transactions was not undertaken.

4.2.5 Hypothesis 5

According to the Hypothesis 5, the higher the level of the investor's protection of the acquirer's country is comparing to the target company's country, the more positive is the post-announcement short-term stock performance of the acquirer.

In the [-3, 3] event window models with time fixed effects (which was chosen to be more appropriate following the Hausman test as described in the section 4.2.2), the Difference in Investors' Protection level was found to have a positive effect on acquirer's CARs. Moreover, the coefficients were statistically significant at least at 10% level. The positive increase of 0.16 to 0.21 percentage points acquirer's CARs in case of one unit increase in the difference between the Investors' Protection indices, is relatively high given the CAR [-3, 3] mean of 0.48%. Yet, from the economic perspective, the positive effect is only approximately 0.16 to 0.21 percentage points of the acquirer's CARs may not be perceived as a significant gain arising from such conditions.

Contribution of M&A to the adoption of higher corporate governance standards has been suggested e.g. by Bris et al. (2008) and supported further by Martynova & Renneboog (2008). Higher investor's protection level in the acquirer's country may contribute to higher CARs in case of full takeover and the resulting target nationality change (positive spillover by law) or even in case of a partial takeover (spillover by control). The results obtained in our analysis (presented in Table 9) corroborate both hypotheses described. Upon acquisition with control change (which is our case), the acquirer's corporate governance standards might be imposed on the target, even when the target is not acquired fully. Subsequently, synergies from the deal might also arise thanks to the better management or reduction private benefits of control. Such future value creation is translated into the CARs upon announcement.

Table 9: Hypotheses 4 & 5 Results – Regulatory Factors Differences in Cross-Border Deals – Time Fixed Effects

Table 9 summarizes results of the Hypothesis 4 and 5 measuring the relationship of the differences in the regulatory factors levels in the acquirer and target country and the acquirer's CARs. CARs of the 7-day event window [-3, 3] used as the dependent variable. Respective t-statistics are reported in brackets. The null hypothesis of the random effects model being the appropriate specification was rejected by the Hausman test at 5% significance level for the models A, B, C, D, E and G; and at 10% level for the model F. Significance levels are estimated using the robust standard errors. *, **, *** represent significance at 10%, 5% and 1% level, respectively. Robust standard errors were employed to correct for autocorrelation and heteroskedasticity.

CAR [-3, 3] mean value = 0.0048		CAR [-3, 3] - Time Fixed Effects						
Variable		A	B	C	D	E	F	G
Independent variables								
ΔPI	Difference in Investor's Protection	*0.0016 (1.95)			**0.0018 (1.99)	**0.0016 (2.02)		**0.0021 (2.25)
ΔEC	Difference in Enforcing contracts		0.0003 (0.47)			0.0006 (0.73)	0.0002 (0.29)	0.0012 (1.16)
ΔRQ	Difference in Regulatory Quality			0.0086 (0.51)	-0.0172 (-0.82)		0.0065 (0.37)	-0.0351 (-1.42)
Deal-specific control variables								
DIV	Diversifying deal	0.0004 (0.01)	0.0307 (1.31)	0.0307 (1.00)	0.0078 (0.25)	-0.0076 (-0.21)	0.0298 (0.93)	-0.0017 (-0.05)
Acquirer or target-specific control variables								
ln(AS)	Acquirer's size	-0.0127 (-1.64)	*-0.0207 (-1.91)	**_-0.0198 (-1.96)	*-0.0122 (-1.86)	*-0.0136 (-1.66)	*-0.0203 (-1.86)	**_-0.0137 (-2.04)
STAT	Target's status	0.0675 (1.25)	***0.1451 (2.61)	***0.1448 (2.62)	0.0627 (1.11)	0.0593 (1.03)	***0.1437 (2.61)	0.0399 (0.58)
RU	Regulated Utilities	***0.0685 (2.48)	***0.1094 (2.49)	***0.1039 (2.49)	***0.0709 (2.84)	***0.0701 (2.47)	***0.1058 (2.37)	***0.0768 (3.03)
AL	Acquirer's leverage	-0.0006 (-0.01)	0.0616 (0.7)	0.0635 (0.65)	0.0025 (0.03)	-0.0186 (-0.23)	0.0596 (0.66)	-0.0333 (-0.37)
Country-specific control variables								
GDP_G	GDP Growth in the target country	***0.0149 (3.87)	***0.0216 (3.57)	***0.0209 (3.88)	***0.0141 (4.22)	***0.0158 (3.69)	***0.0213 (3.63)	***0.0153 (3.94)
HI	Hofstede's Cultural Index	-0.0015 (-1.27)	***_-0.0034 (-3.45)	***_-0.0034 (-3.49)	-0.0012 (-1.12)	-0.0016 (-1.43)	***_-0.0034 (-3.56)	-0.0012 (-1.21)
ML (-3, 3)	Market liquidity	**_-5.0033 (-1.97)	-0.3582 (-0.16)	-0.8471 (-0.43)	**_-4.715 (-2.04)	**_-5.1238 (-2.03)	-0.7203 (-0.35)	***_-4.6751 (-2.23)
	Constant	0.0664 (0.72)	0.0243 (0.23)	0.0216 (0.19)	0.0501 (0.49)	0.0959 (0.98)	0.0283 (0.26)	0.0973 (0.85)
	n	310	310	310	310	310	310	310
	Adjusted R2	77.56%	70.36%	70.39%	78.13%	78.03%	70.35%	79.94%
	F-statistic	11.00	9.54	12.09	9.34	12.33	10.78	14.17

4.2.6 Hypothesis 6

Hypothesis 6 assumed a negative relationship between the increasing difference between the political factor levels in the acquirer and target country, the acquirer's stock performance upon announcement in terms of CARs. Full results of the both random and fixed effects models are presented in the Table 11. The null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 5% significance level for the models A and B (where only Difference in Control of Corruption or Political Stability are included as independent variable, respectively) and at 10% level for the C models (including both variables). Therefore, for the sake of clarity, only fixed effects models are commented on in the following paragraphs. The

fixed effects model yielded inconclusive results in terms of signs of the two indices included. However, none of the models tested showed the coefficients as significant, which poses limitations on the inference regarding the effects of differences in political factors levels on acquirer's CARs upon the announcement of the deal.

Difference in Control of Corruption appeared to have a statistically insignificant positive effect on CARs of 0.8 and 0.74 percentage points per one unit increase in the difference, according to the Model A and C, respectively. This cannot be considered economically very significant, however, comparing to the mean CARs value in the [-3, 3] event window of 0.48%, it represents a significant portion of the variables' effects. Such a positive effect of the difference in Control of Corruption suggests possible exploitation of the market which is ranked lower in terms of this factor despite the risks of such an environment.

Difference in Political Stability was not found significant in case of any fixed effects models. The coefficient signs of this variable were inconclusive. The expected effects on CARs were found both positive of 0.89 percentage points and negative of 3.13 percentage points related to 1 unit increase in the difference in the models B and C, respectively. The latter is the only coefficient we might consider also as an economically significant loss. Overall, not considering the insignificance, I conclude that the differences in political factors levels have a rather positive effect on CARs, since majority of the coefficients obtained were positive. Yet, the inference on acquirer's CARs is, however, statistically limited.

The results suggesting the positive effect of the Control of Corruption on acquirer's CARs variable might provide support for the opportunity to "grease the wheels" mentioned by Dodge & Barkdull (2009) by enabling quicker expansion of the business in an environment where bribery is easier to undertake. On the other hand, as Dodge & Barkdull (2009) suggest, the lower level of political instability may impact negatively the levels of foreign direct investments in the respective country. I also assume that this would be priced in the market and affect the CARs in a negative way. Yet, if business opportunities expected from such acquisitions were to outweigh the risks that are associated with the environment of lower political stability and a possibly higher level of widespread corruption, the effect on CARs would be positive. Given the inconsistency in the coefficient signs and since none of the coefficients in the preferred time fixed effects models were significant, neither of these conclusions can be made and our Hypothesis 6 is not verified.

Further analysis of diversifying and non-diversifying transactions was not carried out, as multiple variables exhibited multicollinearity in the respective subsamples.

Table 10: Hypothesis 6 Results – Political Factors Differences in Cross-Border Deals

Table 10 summarizes results of the Hypothesis 6 measuring the relationship of the differences in the political factors' levels in the acquirer and target country and the acquirer's CARs. CARs of the 7-day event window [-3, 3] used as the dependent variable. Respective z-scores for the random effects regressions and t-statistics for the fixed effects regressions are reported in brackets. The null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 5% significance level for the models A and B and at 10% level for the C models. Significance levels are estimated using the robust standard errors. *, **, *** represent significance at 10%, 5% and 1% level, respectively. Robust standard errors were employed to correct for autocorrelation and heteroskedasticity.

CAR [-3, 3] mean value = 0.0048		CAR [-3, 3] - Random effects			CAR [-3, 3] - Fixed Effects		
Variable		A	B	C	A	B	C
Independent variables							
ΔCC	Difference in Control of Corruption	**0.0099 (2.01)		0.0124 (1.56)	0.0080 (0.64)		0.0074 (0.18)
ΔPS	Difference in Political Stability		0.0076 (1.39)	-0.0036 (-0.41)		0.0089 (0.44)	-0.0313 (0.01)
Deal-specific control variables							
DIV	Diversifying deal	-0.0070 (-0.71)	-0.0062 (-0.63)	-0.0076 (-0.76)	0.0310 (1.13)	0.0332 (1.31)	0.0311 (1.11)
Acquirer or target-specific control variables							
ln(AS)	Acquirer's size	-0.0013 (-0.43)	-0.0013 (-0.43)	-0.0013 (-0.42)	*-0.0202 (-1.95)	-0.0185 (-1.55)	-0.0200 (-1.08)
STAT	Target's status	0.0038 (0.23)	0.0034 (0.2)	0.0028 (0.17)	***0.1516 (2.99)	***0.1526 (3.19)	***0.1517 (3.31)
RU	Regulated Utilities	*0.0208 (1.85)	*0.0213 (1.89)	*0.02 (1.79)	***0.1052 (2.38)	**0.1022 (2.01)	*0.1049 (1.7)
AL	Acquirer's leverage	0.0549 (1.46)	0.0559 (1.47)	0.0545 (1.45)	0.0716 (0.76)	0.0585 (0.53)	0.0705 (0.46)
Country-specific control variables							
GDP_G	GDP Growth in the target country	**0.003 (2.26)	***0.0032 (2.38)	**0.0029 (2.17)	***0.0217 (3.84)	***0.021 (3.86)	***0.0216 (2.74)
HI	Hofstede's Cultural Index	-0.0003 (-1.04)	-0.0002 (-0.65)	-0.0003 (-1.02)	***-0.0034 (-3.61)	***-0.0033 (-3.6)	***-0.0034 (-2.79)
ML (-3, 3)	Market liquidity	-0.1837 (-0.59)	-0.1547 (-0.49)	-0.1809 (-0.58)	-0.6948 (-0.32)	-1.1432 (-0.39)	-0.7389 (-0.15)
	Constant	-0.0218 (-0.75)	-0.0276 (-0.95)	-0.0203 (-0.69)	0.0111 (0.11)	0.0082 (0.09)	0.0109 (0.11)
n		310	310	310	310	310	310
Adjusted R2		19.04%	20.73%	18.11%	70.55%	70.49%	71.32%

4.3 Control Variables Discussion

Ten (and in case of the cross-border subsample nine) control variables were employed in our model in order to capture and disentangle the plethora of effects that arise upon the acquisition announcement. In terms of signs of the coefficients, vast majority of the variables were supporting previous research. Coefficient signs, however, sometimes differed for the random effects and fixed effects model. Since the fixed effects models were generally preferred in case of the regressions with higher explanatory power and significant independent variables, we further comment on fixed effects models mainly. As for the significance and magnitude, unsurprisingly given the previous papers written in the field, majority of the control variables employed (especially target- or acquirer-specific ones) corroborated the results were expected (see the overview in the Appendix B).

The group of deal-specific control variables consisted of dummies controlling for diversifying deal and a cross-border status of the deal. The results obtained suggest that diversifying deals have (often even statistically and economically significant) positive effect on acquirer's CARs, possibly thanks to the co-insurance effect which may arise from such deals. Apart from the random effects models testing the political factors' level differences in acquirer and target countries, coefficients of this variable were positive. However, the random effects model was rejected by Hausman model. Furthermore, cross-border deals mostly appear to contribute to a negative effect on CARs. It proved to be statistically significant at 10% level in the model A testing the effects of political factors in the full sample. This would dispute the positive impact of diversifying deals (in this case we consider diversification in a geographical sense). Negative sentiment arising from the possibly unknown practices in the environment or lack of business network in the target country might explain this effect. Overall, both effects on CARs is in line with my expectations as presented in the earlier sections.

The set of acquirer or target-specific control variables included acquirer's size (measured by logarithm of total assets value), target public vs. private status, dummy controlling for regulated utility sector and a control for acquirer's risk in terms of leverage. Negative coefficients for the acquirer's size (in some cases statistically significant) are consistent with the previous research (Moeller et al., 2004) suggesting that larger acquirers tend to overpay or they may face difficulties capturing the synergies assumed prior the acquisition. Models testing the impact of regulatory and political factor differences also showed regulated utilities control as significant, mostly at 1% level. The positive impact that the regulated status has on the CARs may be a result of positive perception of lower risk due to the certainty imposed by the regulation despite the limited opportunity to increase profit in such companies. Mostly negative (yet statistically insignificant) coefficients for the acquirer's leverage are consistent with the previous research, as the acquisition made by highly leveraged companies imposes a certain risk of default or eventually financial distress. The dummy variable controlling for the public status of the target yielded mostly positive, yet statistically insignificant results. This can be attributed to the fact that private companies are associated with the illiquidity discount resulting in higher CARs upon the announcement (Fuller et al., 2002). Apart from the dummy variable controlling for the regulated status of the company, the obtained results corroborate our assumptions.

Finally, target country GDP growth, Hofstede's Cultural Index level and market liquidity (in terms of stock's average turnover ratio in the event window days) were included as country-specific control variables. GDP growth yielded inconclusive results both in terms of signs and

statistical significance. I assume, that the economic development in the target countries might have been partially captured by the time dummies. Therefore, we cannot conclude whether acquisitions of targets in rapidly growing or developing economies are perceived positively, e.g. due to promising development in terms of the potential of capturing of a large market share in an absolute figure; or negatively, for instance, due to an expected future economic slowdown or decrease in GDP growth. Hofstede's cultural distance index was also found statistically significant mostly in case of models testing the effects of differences between the countries (Hypothesis 4, 5 and 6). Signs of this control variable were not conclusive in the regressions tested, yet they were mostly found negative. Such a negative reception by the market suggests apprehension of potential problematic cultural integration or assimilation. Coefficients of the market liquidity measure were inconclusive in terms of signs although mostly significant for the models tested. Therefore, the inference is limited regarding whether markets with higher liquidity contribute to higher CARs due to more frequent trading possibilities or (on the contrary) limit the CARs effect as frequent trading smoothens out the possible effect. Generally, we conclude that the country-specific control variables yielded mostly inconclusive results in terms of their effects on acquirer's CARs, yet they were necessary to be included in order to disentangle effects of the background and environment surrounding the transactions.

4.4 Robustness Check

In order to analyze the robustness of our results, we implemented several tests and alternative specifications. As described in the section 4.1, we firstly tested for autocorrelation and heteroskedasticity and subsequently employed robust standard errors. As our analysis was undertaken on $[-3, 3]$ representing a midway in terms of event window length, we tested our hypothesis across alternative event windows $[-5, 5]$, $[-5, 0]$ and $[-3, 0]$. Moreover, since more stringent regulation regarding transactions was implemented in the recent years, resulting in an empirically lower run-up in CARs due to leakage, the 3-day event window $[-1, 1]$ included as a robustness check comprises a limited timeframe of deals, specifically only from 2015 on. Finally, the indicator variables approach was further employed to test the robustness of results regarding the effects of differences in the regulatory factors' levels. The obtained results are discussed in the following paragraphs.

Results of the robustness check of Hypotheses 1 and 2 corroborated the previously discussed results. Testing was carried out for the event windows of $[-5, 5]$, $[-5, 0]$, $[-3, 0]$ and the subsample of transactions completed from 2015 and later with the event window of $[-1, 1]$. The statistically significant results were fully supporting the previously discussed results. Majority

of the models with significant coefficients exhibited the most significance at mostly 5% level for the Investors Protection level and Regulatory Quality level for the full sample and Enforcement of Contracts for the subsample of diversifying mergers. The subsample of non-diversifying mergers exhibited significant coefficients results only for Enforcing Contracts variable in the event window [-5, 5], where it was found significant at 10% or 5% level. In terms of the political factor levels, Control of Corruption was found significant at 5% in model apart from the [-5, 5] event window. Our results for Hypotheses 1 and 2, therefore, overall appear consistent.

Furthermore, the robustness check for Hypothesis 3 comparing the CARs of public and private companies with respect to the level of investors' protection in the target country was tested carried out through testing the hypothesis across the previously mentioned event windows. Signs of the obtained coefficients corroborated the previously obtained results for the [-3, 3] event window. Targets in countries lower investors' protection resulted in higher positive CARs comparing to the ones in countries with higher protection. All event windows showed results significant at 1% level. The [-5, 5] event window average CARs for private targets in low investors' protection countries of 2.06% can be also considered as economically significant. Other event windows exhibited CARs of similar magnitude which further supports the initial results.

Results of the robustness check for Hypotheses 4 and 5 overall corroborated the previously found positive effect of the differences in the regulatory factor levels on acquirer's CARs. Apart from the models of event window [-5, 5], the difference in Investors' Protection was significant in all the models at 5% level at least. Majority of the models confirmed the difference in Enforcing Contracts significant at 10% or 5% level. Difference in Regulatory Quality was found significant in a substantial part of the models. It was the only independent variable changing coefficient signs, yet the suggested negative effect was neither statistically not economically significant. The alternative event window [-1, 1] was not tested due to a very limited subsample size (76 observations) for deals between 2015 and 2017 and multicollinearity detected in the regressions. Overall, the results obtained in the robustness check confirm previous findings supporting the positive spillover by law hypothesis with an additional higher frequency of Enforcing Contracts variable significance.

Hypothesis 6 investigating the effect of differences in political factors was further tested in alternative event windows. The results for the event windows off [-5, 0] and [-3, 0] confirmed the ones obtained for the [-3, 3] event window. A positive effect of differences in the level of

Control of Corruption was found in all models including this variable at 10% level. The [-5, 5] event window corroborated previous results with the same coefficient sign, yet the independent variables were not found significant. Due to detected multicollinearity and the highly limited subsample, the event window [-1, 1] was not tested. Yet the results of the robustness check provide moderate support of the positive effect of these differences supporting the hypothesis of leveraging the less controlled environment for business purposes or at least perception of such possibility reflected in the market.

Furthermore, as Martynova & Reneboog (2008) suggest that possible insignificance of regulatory factors coefficients can be caused by cross-border M&A being beneficial not only for the acquirers themselves but also bidding companies from the lower protection country. In order to disentangle such effect, an indicator-variable approach is suggested. Subsequently, we created dummy indicator variables that would control for possible positive spillover by law effect (acquirer's level of the factor is higher than average, whereas target's level is lower than average) or for bootstrapping hypothesis (acquirer's level of the factor is lower than average, whereas target's level is higher than average). Averages considered were computed across all countries for the respective year. In order to avoid testing regressions with multicollinearity arising from the newly added indicator variables, we investigated the positive spillover by law and bootstrapping hypothesis for the Regulatory Quality only. Respective results are reported in the Table 11. The positive coefficient of the indicator variable Regulatory Quality Improvement (Target) was found significant at 1% for the preferable time fixed effects model of the full sample size. This further supports our previous findings supporting the positive spillover by law hypothesis. The acquisition of companies in countries with poor regulatory quality represents opportunities for improvement generating abnormal returns at the transaction announcement. These are also economically significant as the acquisition of a target in a country with lower than average regulatory quality level contributes to 5.99 percentage points higher CARs. These results were not found significant for the subsample of diversifying deals, however, they were corroborated by regressions with the event window lengths of [-1, 1], [-5, 5], [-5, 0], [-3, 0]. Therefore, we assume that this opportunity of improvement is not perceived as very value-creating when the acquirer is from a different industry than the target, but generally, our results support the positive spillover by law hypothesis. Our results are also highly comparable with the findings of Martynova & Reneboog (2008) not only in terms of conclusions but also the significance of the independent and indicator variables.

Table 11: Regulatory Factor Levels Effects of Acquirers' CARs – Indicator Variables Approach Robustness Check

Table 11 summarizes results of the robustness check regarding the regulatory factor levels in the target country and the acquirer's CARs through an indicator variables approach in order to disentangle the positive spillover by law effect and bootstrapping hypothesis. CARs of the 7-day event window [-3, 3] used as the dependent variable. Respective z-scores for the random effects regressions and t-statistics for the fixed effects regressions are reported in brackets. Regulatory Quality Improvement (Target) dummy variable equals 1 when the acquirer's country Regulatory Quality level is higher than the relevant year's cross-country mean and target's country Regulatory Quality level is lower than the relevant year's cross-country mean, suggesting positive spillover by law potential. Regulatory Quality Improvement (Acquirer) dummy variable equals 1 when the acquirer's country Regulatory Quality level is lower than the relevant year's cross-country mean and target's country Regulatory Quality level is higher than the relevant year's cross-country mean, which is a proxy for bootstrapping effect. The null hypothesis of random effects model as the appropriate was rejected by the Hausman test at 10% significance level for full sample regressions. It was not rejected for the subsample of diversifying deals. Significance levels are estimated using the robust standard errors. *, **, *** represent significance at 10%, 5% and 1% level, respectively. Robust standard errors were employed to correct for autocorrelation and heteroskedasticity.

CARs [-3, 3] mean = 0.0085		Full Sample		Diversifying deals	
Variable		Random Effects	Time Fixed Effect	Random Effects	Time Fixed Effect
Independent variables					
RQ	Regulatory Quality	-0.0089 (-0.84)	0.0174 (0.95)	-0.0252 (-1.53)	-0.0015 (-0.06)
Indicator Variables					
RQI_T	Regulatory Quality Improvement (Target)	0.0174 (1.32)	***0.0599 (2.61)	0.0335 (1.47)	0.0656 (1.25)
RQI_B	Regulatory Quality Improvement (Acquirer)	-0.0232 (-0.73)	0.0087 (0.07)	-0.0176 (-0.42)	-0.1418 (-0.76)
Deal-specific control variables					
DIV	Diversifying deal	0.0065 (0.85)	***0.0515 (3.04)		
CB	Cross-border deal	-0.0001 (-0.02)	-0.0068 (-0.35)	-0.0006 (-0.05)	0.0254 (0.60)
Acquirer or target-specific control variables					
ln(AS)	Acquirer's size	***-0.0079 (-3.28)	***-0.0133 (-2.65)	***-0.0085 (-2.44)	-0.0091 (-0.8)
STAT	Target's status	-0.0009 (-0.07)	0.0123 (0.47)	-0.0114 (-0.39)	**0.1426 (2.07)
RU	Regulated Utilities	0.0042 (0.59)	0.0006 (0.04)	0.0034 (0.30)	0.0229 (0.78)
AL	Acquirer's leverage	0.0140 (0.56)	-0.0282 (-0.61)	0.0329 (0.86)	0.0824 (1.12)
Country-specific control variables					
GDP_G	GDP Growth in the target country	**0.0018 (1.95)	*-0.0049 (-1.63)	0.0016 (0.91)	0.0055 (0.41)
HI	Hofstede's Cultural Index	**0.0009 (-1.91)	0.0006 (0.76)	**0.0016 (-2.02)	-0.0020 (-1.22)
ML (-1 1)	Market liquidity	0.5991 (1.27)	**1.2032 (2.11)	0.3181 (0.58)	*-2.8915 (-1.89)
	Constant	**0.0910 (2.00)	0.0248 (0.36)	*0.1539 (1.89)	-0.0261 (-0.2)
n		1106	1106	552	552
Adjusted R2		5.66%	12.80%	0.45%	24.13%

5. Conclusion

5.1 Summary of empirical results and practical implications

In this master thesis, we examined the effects of regulatory and political factor levels on the M&A performance in the infrastructure sector. Increasing interest to deploy capital and include this asset class in investors' portfolio mainly due to the stable periodic cash flow is reflected in intense transaction activity. This further enables the companies to benefit from e.g. the economies of scope or contribute to the economic growth (Balász et al., 2009). These factors represent foundation stones for the importance of the thesis research question. Since management teams and investors often claim the regulatory and political aspects to be the determinants of success or failure of the undertaken transaction (Altman, 2005), we perceive importance in investigating deeper the connection between these factors and the outcomes of M&A in the infrastructure sector. Our paper aims at filling in the gap in the research, as to the best of my knowledge, an analysis of specific factors determining the infrastructure M&A outcomes and their magnitude has not been previously published. Furthermore, we included a previously unused variable, namely Enforcing Contracts in the group of the regulatory factors which appeared to be positively correlated with the M&A performance. Intended inclusion of Corruption Perception Index as another new variable was abandoned due to the high correlation. Yet, comparing to the typically used rule of law variable, we used a differently defined regulatory quality variable, as well as index describing the investor's protection comparing to the widely used La Porta et al.'s (1998). Our analysis also investigated further the differences in outcomes given the public status or the targets and levels of investors' protection in the target country with the basis in the analysis by John et al. (2010).

The empirical analysis was carried out through the event study methodology, followed by OLS regression for which the CARs obtained by the event study were used as a dependent variable. Robustness check was carried out through testing of alternative event windows and an indicator variable methodology. Regulatory and political factor indices or the differences between their levels in the target's and acquirer's country formed the set of independent variables following the methodology of Martynova & Renneboog (2008). Furthermore, relevant tests for multicollinearity, heteroskedasticity and autocorrelation presence were performed and subsequently robust standard errors were employed. Our analysis looked deeper also into the differences in diversifying and inter-industry transactions. As a basis of dataset, we retrieved the deals from Thompson One database. Returns for the corresponding acquirers were obtained through stock prices from Datastream database. Moreover, the regulatory and political factor

indices used as independent variables were obtained from World Bank's Databases of Worldwide Governance Indicators and Doing Business Indicators.

Our hypotheses regarding the levels of regulatory and political factors in the countries were supported by the results. In terms of regulatory factors, significantly positive effect on acquirer's CARs was found especially with respect to the Regulatory Quality. As the Investors' Protection level showed negative and much lower influence, we assume that higher protection of investors does not compensate for the lower quality of regulation or rule of law in the target country. Political factors appear to be positively related with the acquirer's CARs based on the significantly positive results for the Control of Corruption and insignificant ones for the Political Stability. Such results show that the knowledge of the target environment important for doing business in the country in general influences the market perception upon the acquisition. Furthermore, our newly suggested regulatory factor variable of Enforcing Contracts was found to have a significantly positive effect on acquirer's CARs in the subsample analysis concerning diversifying and non-diversifying deals. Differences between significance of regulatory and political factors for the diversifying vs. inter-industry deals were detected. Regulatory factors appear to play an important role for diversifying deals, whereas the opposite is true for the political factors. I assume this effect can be explained by different approaches that might be necessary for such transactions. When acquiring a company in a different industry, the bidder firstly needs to become familiar with the regulatory environment that might not be within his previous area of expertise. Acquirers within the industry might be already aware of the regulatory environment and may become more concerned with potentially secondary factors such as the political environment in the target country.

On the other hand, the negative effect of greater differences between regulatory and political factor levels in acquirer and target country was not confirmed. Our results suggest, therefore, that such differences may, in fact, present opportunities for corporate governance improvement, acquisitions in an environment that is not generally perceived as favorable or "greasing the wheels" for business (Dodge & Barkdull, 2009). Clearly, the opportunities to undertake such transactions must outweigh the risks. Furthermore, a separate analysis of the differences in investors' protection level corroborates the previous findings of Martynova & Renneboog (2008) and the positive spillover by law hypothesis. Improved corporate governance standards leading to greater future value creation are reflected in the CARs upon deal announcement.

Finally, our analysis of the CARs and their relationship with the public status of the target company and the investors' protection level in the target country supported the hypothesis implying that private targets yield higher CARs for acquirers in case of lower investors' protection level. We assume that this may be resulted in by the liquidity discount usually placed on the private targets, which can be even greater in case the target country is remarked by the uncertainty for investors in terms of lower protection.

Several implications arise from our results that overall supported the importance of regulatory and political factors for the M&A performance. Apart from the academic contribution, the results might be of importance to investors, managers or other parties engaged in transactions. Countries with a superior regulatory environment might be more attractive for investors especially in case of diversifying deals. This might motivate countries for higher output quality of their regulatory bodies or enforcement entities in order to e.g. attract capital investments in the country or appeal to more investors (and potentially increase competition in the deal) in case of privatization of certain infrastructure assets. Furthermore, despite the related uncertainty, acquirers might use the opportunity of potentially negotiate higher discount than just the liquidity one on private companies in countries with lower investors' protection level.

5.2 Limitations and future research

This master thesis faces several limitations. Firstly, a substantial part of the dataset analyzed consists of regulated utility companies. Such type of companies does not represent a market for mergers and acquisitions itself due to the limited profit such companies offer. We try to control for this fact by a simple dummy variable distinguishing regulated and unregulated infrastructure, yet due to the diversity of the countries and regulations of the specific infrastructure subsectors, potential inaccuracy may arise.

Secondly, due to the limited data availability from the Thompson One database, the commonly used control variables such as the method of payment and deal size had to be omitted, as they would be too limiting for our dataset. Since larger transactions might result in smaller returns for the acquirers (Alexandridis et al., 2013), cash payment increases the returns of the acquirer especially comparing to the stock-financed deals (Loughran & Vijh, 1997), omission of these control variables may result in an inaccuracy in the coefficients obtained for the independent and control variables used. Similarly, the limited availability of the data regarding e.g. earnings announcements and other corporate events related to the acquirer did not allow us to completely exclude all possible confounding events with the transaction announcements.

Thirdly, a potential limitation can be stemming from the indices used as the independent variables. Their methodology is based on standardized scenarios for a standardized business to ensure the comparability across the global set of economies. Yet, the standardized business includes the assumption of the location in the largest city in the respective economy. In the case of larger countries, potential differences even within the country may arise. However, it is not feasible to collect relevant data to create index scores on the sub-national level. Especially given the large scale of operations of the infrastructure sector companies, I believe this may induce certain inaccuracies in the models used for analysis.

Fourth, the market model used for the event study measures the divergence of the actual return from the expected return. In the ideal case, the indices representing the market benchmarks should be of both respective industry and country, which was however not achievable to match for all countries included.

Fifth, it is important to recognize that choosing a cross-border acquisition instead of domestic one is not exogenous, therefore, endogeneity might influence the results we obtain. According to Rossi & Volpin (2004), acquirers and targets of higher investors' protection level countries are more prone to engage in domestic transactions, rather than cross-border ones. Synergies arising from the transaction involving a target country with a lower investors' protection level need to be significantly greater than the costs associated with subsequent company integration for the bidder to undertake such acquisition. Therefore, a positive relationship between investors' protection levels and stock returns upon the announcement is assumed. The sample-selection bias for the Hypothesis 4, 5 and 6 could be control for example through Heckman's (1976, 1979) procedure which uses a Probit analysis to estimate the probability of executing either a cross-border or a domestic deal.

Some of the shortcomings mentioned would be ideally addressed by future research. A different database availability that future researchers might have could solve the issue with limited data on methods of payment, deal value or corporate events that might be confounding. Thus, additional effects could be captured and the accuracy of the analysis increased.

Furthermore, an extension of the current paper could implement a comparison of other ways to measure regulatory and political factors such as World Economic Forum's Global Competitiveness Report. Since indices published and developed by different entities rank countries based on different grounds, some of them may be found as better capturing or explaining the effects arising upon the acquisition announcements. As such, using a different methodology would provide us with a further robustness check.

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7. Appendices

Appendix A - SIC Industry Codes and Descriptions of selected industries

Industry SIC Code	Description
4499	Water transportation services, nec
4581	Airports and airport terminal services
4812	Radiotelephone communications
4813	Telephone communications, except radiotelephone
4822	Telegraph and other message communications
4899	Communications services, nec
4911	Electric services
4922	Natural gas transmission
4923	Natural gas transmission and distribution
4934	Natural gas distribution
4925	Gas production and/or distribution
4931	Electric and other services combined
4932	Gas and other services combined
4939	Combination utilities, nec
4941	Water supply
4961	Steam and air conditioning supply
4971	Irrigation systems
499A	Cogeneration, alternative energy sources
4952	Sewerage systems
4953	Refuse systems
4959	Sanitary services, nec

The table summarizes the SIC Codes falling within the category of infrastructure sector based on the NAICS Association 2018 websites.

Appendix B – Control Variables Description

Appendix B summarizes the notations and definitions of the control variables used in the models for testing Hypotheses 1, 2, 4, 5 and 6. Furthermore, the expected effects in terms of coefficient sign for the effect on acquirer's CARs is included.

Overview of control variables

Denotation	Name	Type	Definition / Value	Description	Expected sign - acquirer
Panel A: Deal-specific control variables					
DIV	Diversifying deal	Dummy	1 = Same industry subgroup 0 = Different industry subgroup	Deal within the same industry subgroup is defined as the one where both target and the acquirers SIC codes belong to the same core industry group (first 2 digits of the industry code are the same)	Positive
CB	Cross-border vr. Domestic deal	Dummy	1 = Domestic deal 0 = Cross-border deal	Distinguished between a deal where the target is in the same country as the acquirer or not	Negative
Panel B: Acquirer and target control variables					
AS	Acquirer's size	Continuous	Log (Total Assets)	Logarithm of acquirer's Total Assets	Negative
STAT	Target status	Dummy	1 = Private 0 = Public	Public or private status of the target	Positive
RU	Regulated Utility Sector	Dummy	1 = Regulated Utility 0 = Non-regulated Infrastructure Sector	Target belongs to the regulated utility sector, target's SIC industry code begins with 49	Negative
AL	Acquirer's leverage	Continuous	Total Liabilities / Total Assets	Ratio of the acquirer's Total Liabilities and Total Assets	Negative
Panel C: Country-specific control variables					
GDP_G	GDP growth in the target's country	Continuous	GDP growth	GDP growth of the target country in the acquisition-respective year	Positive
HI	Cultural distance	Continuous	Combined Hofstede's index	Euclidian distance in individualism, uncertainty avoidance and long-term orientation index in the target country	Negative
ML	Market liquidity	Continuous	Share turnover ratio	Average trading volume of the respective acquirer's stock divided by the total number of shares outstanding in the event window days	Positive
Panel D: Fixed effects control variables					
T	Time fixed effect	Dummy	1 = Respective day announced 0 = Other day	Value equals to 1 in case of deal announcement on the respective day.	Unsure

Appendix C - Independent and Control Variables Descriptive Statistics – Cross-border Subsample

Appendix C descriptive statistics overview of the independent and control variables for the cross-border transactions subsample used for the regressions with differences in index levels as independent variables.

Descriptive Statistics of Independent and Control Variables - Crossborder Subsample

Variable		Obs	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Independent variables								
Δ PI	Difference in Investor's Protection	310	-1.8049	20.8871	-66.6600	66.6600	-0.1743	3.7302
Δ EC	Difference in Enforcing contracts	310	1.4057	15.0394	-49.3300	41.5800	-0.1076	3.7767
Δ RQ	Difference in Regulatory Quality	310	0.1802	0.8443	-2.2213	2.5348	0.1386	4.4090
Δ CC	Difference in Control of Corruption	310	0.3356	1.0812	-2.7414	3.3423	-0.0113	3.8502
Δ PS	Difference in Political Stability	310	0.2027	0.8734	-2.5651	2.6129	-0.1892	3.3599
Deal-specific control variables								
DIV	Diversifying deal	310	0.4742	0.5001	0.0000	1.0000	0.1034	1.0107
Acquirer or target-specific control variables								
ln(AS)	Acquirer's size	310	6.9503	2.8945	-3.9120	12.0881	-0.4563	3.0579
STAT	Target's status	310	0.8968	0.3047	0.0000	1.0000	-2.6082	7.8026
RU	Regulated Utilities	310	0.5194	0.5004	0.0000	1.0000	-0.0775	1.0060
AL	Acquirer's leverage	310	0.5411	0.2005	0.0188	0.9931	-0.2858	2.4832
Country-specific control variables								
GDP_G	GDP Growth in the target country	310	2.8574	3.3879	-5.6189	25.5573	1.3835	10.5705
HI	Hofstede's Cultural Index	310	49.8516	18.5175	11.0000	95.0000	0.5669	2.3982
ML (-5, 5)	Market liquidity in the event window (-5, 5)	310	0.0061	0.0165	0.0000	0.2497	11.1724	158.2039
ML (-3, 3)	Market liquidity in the event window (-3, 3)	310	0.0066	0.0196	0.0000	0.3012	11.6914	168.4074
ML (-1, 1)	Market liquidity in the event window (-1, 1)	310	0.0070	0.0206	0.0000	0.3074	10.9071	151.0720
ML (-5, 0)	Market liquidity in the event window (-5, 0)	310	0.0061	0.0194	0.0000	0.3081	12.7182	193.4598
ML (-3, 0)	Market liquidity in the event window (-3, 0)	310	0.0065	0.0231	0.0000	0.3751	13.4609	210.2465

Appendix D – Correlation Matrices

Correlation Matrices - Simple Index Regressions

The following matrices provide correlation scores for the independent and control variables in the regressions used to test the Hypotheses 1 and 2 regressing CARs on the regulatory and political factors indices.

	<i>PI</i>	<i>EC</i>	<i>RQ</i>	<i>STAT</i>	<i>CB</i>	<i>DIV</i>	<i>GDP_G</i>	<i>HI</i>	<i>AS</i>	<i>AL</i>	<i>ML</i>	<i>RU</i>
PI	1.0000											
EC	0.0949	1.0000										
RQ	0.5772	0.4048	1.0000									
STAT	-0.0678	0.0255	-0.0278	1.0000								
CB	-0.0636	-0.0044	-0.1121	-0.0750	1.0000							
DIV	0.0015	0.0157	0.0050	-0.1226	0.0373	1.0000						
GDP_G	-0.2351	-0.1277	-0.4026	0.0482	0.0008	-0.0953	1.0000					
HI	-0.5071	-0.1844	-0.8562	0.0401	0.0939	0.0064	0.3698	1.0000				
AS	0.0151	-0.0276	-0.0264	-0.2137	-0.1411	0.2033	-0.1187	0.0217	1.0000			
AL	0.1181	-0.0980	0.0983	-0.0852	0.0021	0.1422	-0.0943	-0.1202	0.3188	1.0000		
ML	0.0047	0.0828	-0.0619	-0.0050	0.0299	-0.0796	0.0705	0.0441	-0.0760	-0.0434	1.0000	
RU	-0.0924	-0.1215	-0.1546	0.0102	0.0505	-0.0220	0.0313	0.1111	0.1082	0.0922	-0.0667	1.0000
VIF Score	1.5581	1.4533	5.9418	1.0654	1.0565	1.0655	1.2240	4.2463	1.2561	1.1658	1.0352	1.0472

	<i>CC</i>	<i>PS</i>	<i>STAT</i>	<i>CB</i>	<i>DIV</i>	<i>GDP_G</i>	<i>HI</i>	<i>AS</i>	<i>AL</i>	<i>ML</i>	<i>RU</i>
CC	1.0000										
PS	0.8592	1.0000									
STAT	-0.0278	-0.0467	1.0000								
CB	-0.1036	-0.0889	-0.0750	1.0000							
DIV	-0.0208	-0.0093	-0.1226	0.0373	1.0000						
GDP_G	-0.3407	-0.3418	0.0482	0.0008	-0.0953	1.0000					
HI	-0.8538	-0.7256	0.0401	0.0939	0.0064	0.3698	1.0000				
AS	-0.0571	-0.0181	-0.2137	-0.1411	0.2033	-0.1187	0.0217	1.0000			
AL	0.0685	0.0585	-0.0852	0.0021	0.1422	-0.0943	-0.1202	0.3188	1.0000		
ML	-0.0698	-0.0713	-0.0050	0.0299	-0.0796	0.0705	0.0441	-0.0760	-0.0434	1.0000	
RU	-0.1488	-0.0979	0.0102	0.0505	-0.0220	0.0313	0.1111	0.1082	0.0922	-0.0667	1.0000
VIF Score	6.7981	3.8625	1.0636	1.0513	1.0659	1.1998	3.8197	1.2587	1.1442	1.0166	1.0458

Correlation Matrices - Difference-based Independent Variables

The following matrices provide correlation scores for the independent and control variables in the regressions used to test the Hypotheses 5, 5 and 6 regressing CARs on the regulatory and political factors index differences.

	ΔPI	ΔEC	ΔRQ	<i>STAT</i>	<i>DIV</i>	<i>GDP_G</i>	<i>HI</i>	<i>AS</i>	<i>AL</i>	<i>ML</i>	<i>RU</i>
ΔPI	1.0000										
ΔEC	0.0920	1.0000									
ΔRQ	0.5056	0.4565	1.0000								
<i>STAT</i>	0.0144	0.0586	0.0016	1.0000							
<i>DIV</i>	-0.0506	0.0209	0.0306	-0.1449	1.0000						
<i>GDP_G</i>	0.1819	0.0660	0.2889	0.0257	0.0234	1.0000					
<i>HI</i>	0.2401	0.1378	0.4814	0.1057	0.0925	0.3465	1.0000				
<i>AS</i>	-0.1351	0.0559	-0.0696	-0.2151	0.1982	-0.1449	-0.0531	1.0000			
<i>AL</i>	-0.1442	0.1432	-0.0895	-0.0725	0.1407	-0.1220	-0.0313	0.3580	1.0000		
<i>ML</i>	0.0744	0.0137	0.1179	0.0025	-0.0992	0.1024	0.0795	-0.1103	-0.1811	1.0000	
<i>RU</i>	0.0559	0.0803	0.0270	-0.0505	-0.0174	-0.0614	0.0324	0.1929	0.2361	-0.1423	1.0000
VIF Score	1.3784	1.3464	2.1409	1.0578	1.0575	1.1562	1.3998	1.2234	1.2284	1.0367	1.0757

	ΔCC	ΔPS	<i>STAT</i>	<i>DIV</i>	<i>GDP_G</i>	<i>HI</i>	<i>AS</i>	<i>AL</i>	<i>ML</i>	<i>RU</i>
ΔCC	1.0000									
ΔPS	0.8086	1.0000								
<i>STAT</i>	0.0040	-0.0001	1.0000							
<i>DIV</i>	0.0438	-0.0101	-0.1449	1.0000						
<i>GDP_G</i>	0.2434	0.1561	0.0257	0.0234	1.0000					
<i>HI</i>	0.4543	0.3376	0.1057	0.0925	0.3465	1.0000				
<i>AS</i>	-0.0455	-0.0598	-0.2151	0.1982	-0.1449	-0.0531	1.0000			
<i>AL</i>	-0.0492	-0.0943	-0.0725	0.1407	-0.1220	-0.0313	0.3580	1.0000		
<i>ML</i>	0.1264	0.0961	0.0025	-0.0992	0.1024	0.0795	-0.1103	-0.1811	1.0000	
<i>RU</i>	0.0217	-0.0010	-0.0505	-0.0174	-0.0614	0.0324	0.1929	0.2361	-0.1423	1.0000
VIF Score	3.2258	2.8670	1.0490	1.0635	1.1530	1.3587	1.2183	1.1937	1.0447	1.0681